

Reduplication Without Syllable Copying*

Abstract

Several theories of reduplication have attempted to accommodate the typologically rare option of syllable copying, arguably attested in Yidj (*mu.la-mu.la.ri* ‘initiated men’, *ŋa.lal-ŋa.lal* ‘big ones’). We revisit the pattern, concluding that it does not support the traditional characterization. Rather, Yidj reduplication targets morphological roots most of the time, and truncates roots longer than two syllables after the second vowel. The theory of reduplication best suited for this type of pattern is Serial Template Satisfaction, which we modify to allow only segmental and morphological copying. The shape of templates is mediated partly by prespecification at the UR level and partly by markedness constraints. This theory is compared with two alternatives: Base-Reduplicant Correspondence and Morphological Doubling Theories.

1 Introduction

Moravcsik (1978) identifies a typological generalization about reduplication: the extent of copying does not depend on syllabification in the base. The famous exception is Yidj (Dixon 1977a), whose disyllabic reduplicant has been said to end in V if the second syllable is open and in C if it is closed (*mu.la-mu.la.ri* ‘initiated men’, *ŋa.lal-ŋa.lal* ‘big ones’). To accommodate Yidj, various theories of reduplication have had to incorporate syllable copying devices that do not appear to be motivated elsewhere. What we aim to do in this paper is argue that these devices should be eliminated. We provide a complete analysis of reduplication in Yidj argue that it has no syllable copying.

Our analysis of Yidj is couched in Serial Template Satisfaction theory (STS, McCarthy et al. 2012, Lamont 2023). As the name suggests, this theory assumes a serial architecture for phonology and templatic representations for reduplicative morphemes. We justify both assumptions extensively below. Like other theories of reduplication, STS was originally designed to accommodate syllable copying. Our simplified version of STS has just two kinds of copying: segmental strings and morphological constituents (morphemes, XPs). We argue that this theory incorporates the best insights of preceding work on reduplication without adopting the excesses that have been implicated in overprediction.

The relevance of Yidj to reduplication theory was first noticed by Nash (1979; 1980) and Marantz (1982). Marantz’s Copy-and-Association theory of reduplication aims in part to derive Moravcsik’s Generalization. Rather than viewing reduplication as a transformation rule, as in older rule-based approaches, Copy-and-Association theory builds on McCarthy (1979) in assuming that reduplicative morphemes have a shape that is pre-determined in the UR, as C/V templates. The templates are filled by autosegmental association, duplicating the base’s segmental content and linking it to the C/V template. We illustrate this on Tohono O’Odham, where the shape of the reduplicant is more or less invariant, CV- (see (1)). The derivation first copies the base string, *maihogi-maihogi*, and then proceeds to link segments left-to-right to compatible C or V slots, i.e., [m] to C, and [a] to V. The rest of the string, *ihogi*, is unpronounced:

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(1) Tohono O’Odham CV- reduplication (Hale 1959) in Copy-and-Association theory

Sg	Pl /CV-/	UR:	Association:	SR:
a. maihogi	má-maihogi	‘centipede’	/CV-maihogi/ → CV	→
b. go:ki	gó-goki	‘track’	 maihogí maihogi	[ma-maihogi]

When the reduplicant does vary in shape, Copy-and-Association specifies a cobbled representation in the UR that is the maximal expansion of the reduplicative string. For example, in Dakota, reduplication copies a single syllable, but it varies between -CCV and -CVC: *haska-ska* ‘be tall’, *xap-xap-a* ‘to rustle’. To accommodate this, the reduplicative UR contains a superset string, /-CCVC/. Since association is string-driven and edge-in, *haska-ska* is generated by linking [a] to the rightmost available V slot, while *xap-xap* results from linking [p] to the rightmost available C slot.

It is the superset string assumption that prevents this theory from handling Yidj¹ correctly. In Yidj, the reduplicant does appear to vary based on derived phonological properties of the base. McCarthy et al. (2012, p. 195) characterize its reduplication as follows: “Yidiny copies the first two syllables of the stem, reduplicating σ CV if the second stem syllable is open and σ CVC if it is closed” (cf. (2a–b vs. c–d)). This pattern cannot be analyzed in Copy-and-Association theory: RED must be at least /CVCCVC/ (*nalal-nalal*, *birbi-birbi*)—but such a template predicts **mular-mulari*. Marantz’s solution (1982:454) is to assume a mixed representation where syllables are part of a morpheme’s UR representation, and syllables are what is copied into the reduplicative morpheme. We will see that other theories, such as Morphological Doubling (§5.2), invoke a similar constituent, which has no motivation except to handle reduplication. But Marantz is not satisfied with this solution: “*Why, of all the reduplication processes studied by Moravcsik, myself, and others, is there only one clear example of syllabic reduplication (namely, Yidin’)?*” (Marantz 1982:456).

(2) Yidj as in Marantz (1982) and McCarthy et al. (2012)

a.	bu.na	bu.na-bu.na	‘woman/PL’	σ CV copied
b.	mu.la.ri	mu.la-mu.la.ri	‘initiated man/PL’	
c.	ŋa.lal	ŋa.lal-ŋa.lal	‘big/lots of big (ones)’	σ CVC copied
d.	gin.dal.ba	gin.dal-gin.dal.ba	‘lizard sp./PL’	
e.	ga.lam.ba.ra	ga.la-ga.lam.ba.ra	‘march fly/PL’	homorganic NC clusters are not split
f.	/jimuru/	ji.mu-ji.mur	‘house/PL’	“/r/ is underlyingly an onset”

As can be seen in (2e–f), there are some edge cases in Yidj: thus, homorganic nasal-stop clusters are not split up by reduplication in (2e), and stems with lexicalized apocope are unexpectedly copied as though the last consonant is an onset, as in (2f)). We will devote a fair amount of attention to these and other edge cases, and we will argue that (2d–e) are morphologically missegmented. Furthermore, there is a pattern of reduplication that deviates from the patterns above: in verbs, reduplication optionally copies single -C suffixes, e.g. [wala(l)-wala-l] ‘look thoroughly’. To our knowledge, verbal reduplication has not been analyzed since Dixon (1977a) described it.

Since Marantz’s discussion, the only example of syllable-copying reduplication that has been put forward is Hiaki (Yaqui), but it is controversial due to a high degree of lexical idiosyncrasy (Harley and Florez Leyva 2009; Haugen and Hicks Kennard 2011; McCarthy et al. 2012; Haugen 2014, see §6.1). Yidj has thus remained the best example of syllable copying the literature (McCarthy and Prince 1990; Hicks-Kennard 2004; Haugen and Hicks Kennard 2011; McCarthy et al. 2012; Zukoff 2023; Wang 2024).

¹Yidj is a Pama-Nyungan language originally described by Dixon (1977a); it is sometimes written as Yidin’ and Yidiny. Its current status is unverified (either critically endangered or extinct, according to Ethnologue). Throughout, Dixon’s transcriptions have been converted to the IPA, except [y] stands for a glide.

As we demonstrate, the support for syllable copying in Yidj is scant, resting on just three cases (out of about 100, which we analyze exhaustively in this paper). There are far more cases that suggest a morpheme-based reduplication pattern, which can be informally described as follows:

- (3) **Yidj reduplication**: copy the root if disyllabic; if the root is longer, copy the first σ CV.

The remainder of the article is structured as follows. Our modified version of STS is elaborated in §2. The full Yidj reduplication pattern is in §3, followed by our analysis of nominal (§4, 4.2, and 4.4–4.4) and verbal reduplication (§4.5). In §5 we discuss two alternatives: Morphological Doubling Theory (MDT, Inkelas and Zoll 2005) and Base-Reduplicant Correspondence Theory (BRCT, McCarthy and Prince 1995). We argue that they encounter issues either with the basic nature of the pattern or with the details of the many edge cases, such as the opaque interaction of vowel/zero alternations with reduplication. We also discuss the alternative couched in terms of learnability explanations, inspired by some recent artificial grammar experiments on reduplication (Haugen et al. 2022; Wang 2024, see §5.4). Finally, in §6 we discuss some additional issues, including the other cases that have been analyzed as syllable copying, and role of syllables in infixation. Section 7 concludes.

2 A Theory of Reduplication With Segmental And Morphological Copying

2.1 Templates

Our theory represents reduplicative morphemes as **templates**, which lack some part of segmental structure in the UR. Templates can be prosodic constituents, as in (4), or they can consist of C/V skeleta as in Copy-and-Association.²

- (4) Common reduplicative templates exemplified

$/\sigma_{\mu}-/$	Tohono O’odham (Hale 1959), see (1)		
$/\sigma_{\mu\mu}-/$	Ilokano (Hayes and Abad 1989)		
	kaldín	<u>kal</u> -kaldín	‘goat/+pl’
	púsa	pus-púsa	‘cat/+pl’
	ró?ot	<u>ro</u> :-ró?ot	‘leaves, litter/+pl’
$/[\]_{\text{Ft}}-/$	Yidj (Dixon 1977a), see (2) and §3		
$/[\]_{\text{PwD}}-/$	Kihehe (§5.2.2)		
	kú-gohomóla	kú-gohomola-gohomóla	‘to cough/+a bit’

What makes a morpheme reduplicative is that it (a) has a templatic UR and (b) is supplied with segmental content via copying. An empty template would violate markedness constraints of the HEADEDNESS family, which require prosodic units to have heads (Selkirk 1995). The empty structure can be filled by several operations. In reduplication, this is copying, and in other templatic morphology it can be epenthesis or lengthening/gemination (as in Arabic or Choctaw, see Lombardi and McCarthy 1991 as well as McCarthy 1979 et seq.). Thus, the filling of the reduplicative template is driven by markedness, and it compels a faithfulness violation. In (5) is sketch of the Tohono O’odham example analyzed in STS. The input for the plural of [maihogi] has an empty light syllable prefix in the UR. GEN supplies (at least) three candidates for this input: (5b) has an empty syllable, and is faithful to the input, but it violates HEADEDNESS. Candidate (c) fills the empty syllable with an epenthetic vowel, while the winner, (a), applies a copying operation (defined in §2.3) to fill the empty syllable:

²For other work on reduplication that either assumes or critiques templates, see Gafos (1998); Kirchner (2010); Riggle (2006); Zimmermann (2021), a.o.

(5) Analyzing Tohono O’odham light syllable reduplication in STS: [ma-maihogi] ‘centipedes’

Step 1:	/σ _μ - maihogi/	HEADEDNESS[σ]	DEP-V	*COPY-SEG
a.→	[ma] _σ -maihogi			*
b.	σ _μ -maihogi	*W		L
c.	[ə] _σ -maihogi		*W	L

As we explain in §2.3, the full set of candidates will include some hopeless (harmonically bounded) ones, e.g., [σ_μ-maihogi-ma], which has an empty syllable and puts the copy in the wrong place. A viable candidate puts the copy in a position that achieves template satisfaction.

While some aspects of templates’ shapes are determined in the UR, their filling is modulated by markedness (Steriade 1988, McCarthy and Prince 1994a, McCarthy et al. 2012, a.o.). It is sufficient to specify that the reduplicant is a foot; the details of its appearance can be controlled by the language-specific ranking of markedness constraints (e.g., feet are binary at the syllabic level in Yidj but at the moraic level in Manam, see §6.1). NoCODA ensures that the reduplicant defaults to vowel-finality. Still other aspects of the shape are down to the copying mechanism, as we explain below.

2.2 Harmonic Serialism

STS is couched within Harmonic Serialism, a serial constraint-based framework (McCarthy 2008b; Pruitt 2010, a.o.). In Harmonic Serialism (HS), each derivational step entails generating candidates that differ from the preceding step by one change. The candidate set is evaluated by the constraint ranking, and the winner is submitted to the next derivational step. The derivation ends when the last winner is mapped to itself, i.e., when no further harmonic improvement is possible.

We briefly illustrate by deriving the pattern in American English whereby schwa epenthesis feeds flapping, as in /nat-d/ → [nætəd] ‘knotted’. We will assume two markedness constraints: one against sequences of too-similar coronals *COR-COR, and another that bans alveolar stops between vocoids, *VTV. In the first step, the fully faithful candidate [nat-d] loses to the candidate with schwa epenthesis (6a–b). Note that [nætəd] is not a candidate at this step because it requires two changes from the input. The candidate with schwa epenthesis moves on to the next round, where the viable options are to flap or do nothing. Flapping wins, and no further improvements are possible at step 3 (not shown).

(6) Illustration of a Harmonic Serialism derivation: English flapping

Step 1:	/nat-d/	*COR-COR	*VTV	DEPV	IDENT[son]
a.→	nætəd		*	*	
b.	natd	*W	L	L	

Step 2:	nætəd	*COR-COR	*VTV	DEPV	IDENT[son]
c.→	nætəd				*
d.	nætəd		*W		L

HS permits only local, limited optimization, without global lookahead. HS can therefore derive some opacity effects, just as theories with rule ordering (see, e.g., Elfner 2016 on stress-epenthesis interactions). Continuing with English, for some North American speakers, flapping counterbleeds Canadian Raising (/aɪ/ → [ʌɪ]/ ____ [-voice]), thus [saɪt] ~ [sʌɪrəd] ‘sighted’, [saɪd] ~ [sʌɪrəd] ‘sided’. Since the flap is a sonorant, the diphthong did not need to raise. In HS, this can be analyzed by establishing as a top priority the constraint *aɪ[-voice], penalizing low diphthongs followed by voiceless obstruents. The ranking of markedness constraints dictates the order in which operations apply: diphthong raising in step 1 (7a), then schwa epenthesis between adjacent coronal stops (7d), and finally flapping (7f). If the markedness

ranking were different so that epenthesis applied first, flapping could then simultaneously remove the *aI[-voice] violation and the *VTV violation, without opacity (as in the dialects of speakers who neutralize ‘sighted’ and ‘sided’).

(7) Analyzing counterbleeding opacity in Canadian Raising

Step 1: /saɪt-d/	*aI[-voice]	*COR-COR	*VTV	DEPV	IDENT[son]	IDENT[low]
a. → saɪtd		*				*
b. saɪtəd	*W	L	*W	*W		L
c. saɪtd	*W	*				L

Step 2: saɪtd	*aI[-voice]	*COR-COR	*VTV	DEPV	IDENT[son]	IDENT[low]
d. → saɪtəd			*	*		
e. saɪtd		*W	L	L		

Step 3: saɪtəd	*aI[-voice]	*COR-COR	*VTV	DEPV	IDENT[son]	IDENT[low]
f. → saɪrəd					*	
g. saɪtəd			*W		L	

By contrast, in parallel Optimality Theory, no ranking of these constraints will favor [saɪrəd] over [saɪtəd]: the latter is more faithful. For various arguments in favor of HS, see McCarthy (2008b), Pruitt (2010), Kimper (2011), Elfner (2016), Lamont (2023), and references therein.

In Yidiɲ reduplication, there are two opaque subpatterns, including the interaction between apocope and reduplication in /ʃimuru/ → *ʃimu-ʃimur* ‘houses’. Apocope fails to feed reduplication, so the [r] is unexpectedly not copied despite being root-final on the surface in the base (see §4.4). This is one of the patterns that non-serial theories such as BRCT struggle with.

The main architectural modification in Harmonic Serialism concerns GEN, the candidate-generating component. Instead of having Optimality Theory’s Freedom of Analysis (McCarthy and Prince 1993b), GEN in HS applies one operation per candidate per derivational step. The operations are the domain of faithfulness constraints, which penalize their application rather than track discrepancies between the input and the output via correspondence. For exposition purposes, we will continue using the familiar names of Correspondence-Theoretic (McCarthy and Prince, 1995) constraints (MAX, DEP, IDENT, etc.), but they are formally different. An operation is a function from an input to a set of candidates that differ from the input by one specific change. A faithfulness constraint assigns a violation mark to any candidate generated by an operation.

Take DEP, for example. In classic OT, DEP assigns a violation mark for every segment in one string (the output, or the reduplicant) that lacks a correspondent in another string (the input, or the base). Classic OT’s DEP can assign an unbounded number of violations, since epenthesis can apply an unlimited number of times to any given string. By contrast, in HS, DEP does not need to track correspondence because there can only be one inserted segment per candidate³. Thus, DEPV penalizes candidates generated by the function EPENTHESIS(V, x). Its input is the string x, and its output includes every candidate that adds one V somewhere in the string x. For schwa epenthesis applied to a string of length N, this will generate N+1 candidates, as in (8).

(8) Result of applying EPENTHESIS(V, pat-k):

/pat-k/ əpatk, patək,
 pəatk, patkə
 paət̩k,

³We are sidestepping the issue of epenthetic segment quality; for some pertinent discussion see McCarthy (2008a); Mooney (2023b), a.o.

- (9) DEP(V): Assign one violation mark to any candidate generated by applying EPENTHESIS(V, x) to string x .

Every candidate set obligatorily includes some fully faithful candidates, as well (Moreton 1996/1999; McCarthy 2003a, a.o.), which may differ from each other in markedness violations (e.g., syllabification) but are not the result of any faithfulness-breaching operations.

2.3 COPY-SEG

McCarthy et al. (2012)’s original version of STS includes copying operations targeting contiguous strings of segments, syllables, other prosodic levels, and morphosyntactic structures. We diverge from McCarthy et al. (2012) by excluding COPY-SYLLABLE.⁴ The main reason to omit such operations from the theory is to prevent faithfulness constraints from referring to non-contrastive structure. Syllable structure is never contrastive, though it can vary between languages: thus, *patra* may be syllabified *pat.ra* in one language and *pa.tra* in another, but no language contrasts the two. The usual explanation for this is that languages can rank syllable markedness constraints differently (e.g., NoCODA and *COMPLEXONSET), but no faithfulness constraints refer to syllables (McCarthy 2002, pp. 73–74). Our version of HS GEN routinely builds and rebuilds syllable structure, but the operations are not associated with a faithfulness cost.

McCarthy et al. (2012) do not formalize COPY operations, saying only that “GEN also includes an operation COPY(X) that creates a copy of a string of constituents of type X (with their contents), places that copy anywhere, and integrates it into pre-existing prosodic structure.” (p. 179). We can formalize this more explicitly for string copying as the following function:

- (10) defn. COPY-SEG: Given an input string $x_1 \dots x_n$ where x is a segment,
 COPY: return every copy $y_i \dots y_j$ where $i \geq 1$ and $j \leq n$;
 Concatenate $y_i \dots y_j$ and $x_1 \dots x_n$.
 defn. CONCATENATE: for every a such that $1 \leq a \leq n$,
 return $y_i \dots y_j \mid_{@a} x_1 \dots x_n$ where $\mid$ splits $x_1 \dots x_n$ on a .

This operation does what McCarthy et al. say: copies some contiguous substring of the input, and returns the input plus the copy placed somewhere before, inside, or after. The “before” and “after” options are prefixation and suffixation respectively, and they will be optimal when the reduplicative template precedes (or follows) the input string. The “anywhere” proviso of McCarthy et al.’s informal definition is intended to cover infixing reduplication; we do not discuss this in depth here but it raises some interesting issues that must be explored in later work.

As defined in (10), COPY-SEG can copy any contiguous substring of the input. All the candidates generated by COPY-SEG from an input containing the segments /pati/ are shown in (11). We assume that COPY-SEG ignores the morphological makeup of the input: whether the string contains just one morpheme, /pati/, or more than one, as in /pa-ti/, /pat-i/, etc., all the valid applications of COPY-SEG will look the same. Indeed, a sure diagnostic of segmental (as opposed to morpheme-level) copying is that the copied string can span morpheme boundaries. As with other HS candidate-generating operations, the set of candidates is necessarily finite. COPY-SEG generates 50 candidates given a 4-segment input. In parallel OT, by contrast, the set of candidates is infinite, in part due to GEN’s free application of copying and epenthesis (see McCarthy 2002, 1.1.3).

⁴We also do not treat as a separate derivational step the insertion of an empty syllable when filling in a foot, mostly for convenience (to make tableaux more compact) and because nothing hinges on assuming this operation here.

- (11) Possible single-step applications of COPY-SEG; prefixes and suffixes are underlined and infixes are shown in angled brackets

Input to reduplicate	Valid candidates generated (N=50)	Invalid applications
/pati/, /pa-ti/, /p-ati/	<u>pati</u> -pati, <u>pat</u> -pati, <u>ati</u> -pati, <u>pa</u> -pati, <u>at</u> -pati, <u>ti</u> -pati, <u>p</u> -pati, <u>a</u> -pati, <u>t</u> -pati, <u>i</u> -pati (N=10)	p_ <u>ti</u> , <u>a</u> _i, pa_i, fati, pate, apti, papatiti-pati, etc.
	p<pati>ati, pa<pati>ti, pat<pati>i, p<pat>ati, pa<pat>ti, pat<pat>i, p<ati>ati, pa<ati>ti, pat<ati>i, p<pa>ati, pa<pa>ti, pat<pa>i, p<at>ati, pa<at>ti, pat<at>i, p<ti>ati, pa<ti>ti, pat<ti>i, p<p>ati, pa<p>ti, pat<p>i, p<a>ati, pa<a>ti, pat<a>i, p<t>ati, pa<t>ti, pat<t>i, p<i>ati, pa<i>ti, pat<i>i (N=30)	
	pati- <u>pati</u> , pati- <u>pat</u> , pati- <u>ati</u> , pati- <u>pa</u> , pati- <u>at</u> , pati- <u>ti</u> , pati- <u>p</u> , pati- <u>a</u> , pati- <u>t</u> , pati- <u>i</u> (N=10)	

While the invalid applications in (11) cannot be generated in one step, they can be the result of longer derivational chains. For example, a non-contiguous copy (e.g., *p_{ti}*) will require an additional deletion step after copying. Non-contiguous copying is attested, but languages that have it also have independent markedness requirements that hold of either affixes in general or of phonological contexts in which the reduplicated morphemes appear (see McCarthy et al. 2012 §6.6 on Seediq and Sanskrit). Featural changes (*fati*, *pate*) require subsequent IDENT violations. Along the same lines, reduplication cannot copy parts of a complex segment: /mbati/ cannot be copied as just *bati*, but it can be reduced to [bati] by denasalization.

As with any operation in Harmonic Serialism, segment copying violates a faithfulness constraint, *COPY-SEG.

- (12) *COPY-SEG: “Assign a violation mark for any candidate generated by the operation COPY-SEG.”

*COPY-SEG is a faithfulness constraint that interacts with the rest of the phonological system. In reduplication, violations of *COPY-SEG are compelled by HEADEDNESS (§2.1). Outside reduplication, violations of *COPY-SEG can be compelled by markedness—for example, in copy epenthesis (see Mooney 2023b and references therein). Copying to satisfy a disyllabic minimal word target is attested both in adult languages (Movima, Haude 2006, 3.7.9) and in child speech (Klein 2005).

2.4 COPY-MORPH AND COPY-XP

The COPY-MORPH operation is restricted to copying all the material associated with a morph. Whether an item is a morph or not can be ascertained by consulting the lexicon’s Vocabulary Insertion rules (Halle and Marantz 1993). The operation takes a morph and places a copy of it next to the original:

- (13) COPY-MORPH: “Given the input A . . X . . B, where X is a morph, return A . . X X . . B.”

There is a corresponding faithfulness constraint *COPY-MORPH, defined similarly to *COPY-SEG, which penalizes a single instance of morpheme copying.

In our implementation, Copy-Morph always places the copy next to the original, but reduplication can in principle produce a discontinuous constituent. This happens if infixation follows reduplication—as is the case in multiple Austronesian languages (e.g., Tagalog *t<um>a-takboh* ‘will run’, Marantz 1982:439). Reordering of regular (non-infixing) affixes can also render the reduplicant non-adjacent to its base: in Madurese, /ka-toron-an/ ‘descendant’ is pluralized as [ron-ka-toron-an], with [ron] being separated by the

prefix [ka-] and the stem syllable [to] from the original (Davies 2010:113, see §6.1 for more on Madurese). We assume that in such cases, some post-syntactic affixal movement/repositioning must take place, but we do not analyze these cases as that would take us too far afield. The important point for us is that morphemic copying can only be optimal in the cases we discuss if it fills a reduplicative template. Since copying satisfies HEADEDNESS, putting the copied morph in the slot where the reduplicative affix sits in the structure will be the surest way to fill the template.

Whenever multiple morphemes are copied, we assume that either string-level copying or XP-level copying is taking place. Copying of branching XPs is independently necessary for phrasal reduplication: English contrastive reduplication can copy a full VP (Ghomeshi et al. 2004: “*SIT ON IT sit on it*”). There is a lot more to say about this type of copying, but for now it will suffice to define it as taking all the phonological material inserted to expone the terminals and repeating it (see Marantz 1982; Ghomeshi et al. 2004; Inkelas and Zoll 2005, a.o. and §5.2.2).

- (14) COPY-XP: “Given the input [A B C]_{XP}, where A, B, C are exponents of terminals dominated by XP, return ABC ABC.”

The constraints *COPY-SEG, *COPY-MORPH, and *COPY-XP are independently violable and are not in a stringency relationship. Copying an XP does not entail copying a morph, and copying a morph does not entail copying a segmental string. A consideration of the logical possibilities should clarify why this is the case. A morpheme can be segmentally empty: this is the case for reduplicative affixes. Thus, COPY-MORPH could take /σ-/ as its input and return σσ. A morpheme can also comprise a mix of segments and floating features or abstract stress markings (see Wolf 2007, a.o.). Formally, it is possible to apply either COPY-SEG or COPY-MORPH to a morpheme with the same result: thus, [pati-pati] could be generated by copying the root *pati* or the string *p,a,t,i*. The resulting candidates would have different faithfulness violation profiles based on which operations generated them (much as in Correspondence Theory, where [pati-pati] can be generated by epenthesis of segments [p,a,t,i] that do not correspond to base segments [pati]). Note, however, that it is impossible to, say, copy [pat-pati] using COPY-MORPH. Such a candidate can only be produced via COPY-SEG or via COPY-MORPH with deletion (a MAX-V violation) on a subsequent step.

The reduplicative morphology of Yidiñ, we argue, crucially distinguishes between copying strings that constitute morphemes and strings that are parts of morphemes (see §4.5):

- (15) One morpheme vs. two in Yidiñ

-C affix not copied:	root-final C copied:
a. guyga-guyga- <u>m</u> ‘in the evening’	a’. giram-giram ‘cramp’
b. bulba-bulba-l ‘ground a lot’	b’. ɲalal-ɲalal ‘lots of big ones’
c. gaʒi-gaʒi-n̄-bara ‘redup-long.way-gen-abs’	c’. guran-guran ‘tall pl.’

2.5 Morphological Assumptions

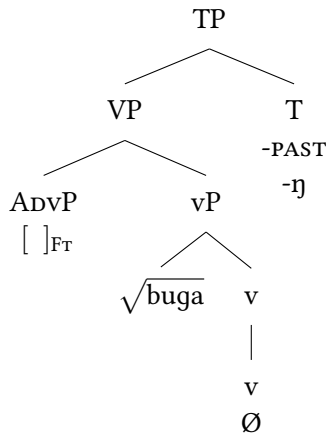
Separation. An important assumption about the morphology-phonology interface for us is that phonology gets access to the entire string created by Vocabulary Insertion, rather than getting it in cyclic chunks (see Chomsky and Halle 1968; Kiparsky 1982 and many others). We will see several cases where it is necessary to assume that reduplicative copying gets access to segments from morphemes that it should not be able to access if phonological derivation proceeded in a strictly cyclic manner—these are classic **bracketing paradoxes**. These paradoxes arise in verbal reduplication in Yidiñ (§4.5), Kihehe (§5.2.2), and Madurese (§6.1). While such paradoxes have long been known in reduplication (Marantz 1987), they receive no explanation in theories that chunk phonological derivations by morphosyntactic cycles such as phases (Marvin 2002, Sande et al. 2020, Fenger et al. 2025, a.o.). Thus, in our theory, phonology receives

as input the full linearized string of morphs, along with the bracketing/hierarchical structure (much as in the original SPE). To the extent that hierarchical structure plays a role in phonological phenomena, as in cases where branching morphosyntactic constituents are copied, the morphosyntactic structures may be referenced by phonological constraints, but morphosyntactic and phonological structures do not have to be coextensive (on this, see McCarthy and Prince 1993a; Selkirk 1995; Benua 1997; Truckenbrodt 1999, and many others). This is Separation.

To briefly illustrate, consider the Yidij verbal reduplicated form [bugaŋbugaŋ] ‘eats fast’. The present tense suffix /-ŋ/ is structurally higher than the adverbial reduplicative prefix, as we explain in §4.5—and yet the reduplicative prefix copies it. The assumption is that the structure submitted to phonology as input is a linear string, /[]_{Ft} -buga-ŋ/, with all the morphemes available at once rather than being constrained by the bracketing structure. The example is not unique: reduplicative morphemes can copy both above and below their syntactic scope (see §5.2.2, §6.1).

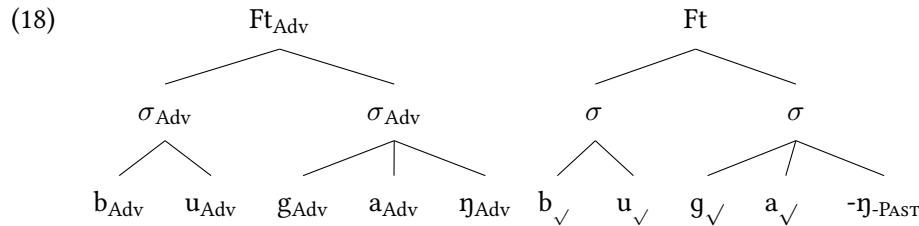
(16) Linear vs. hierarchical structure in Yidij

/[]_{Ft} -buga-ŋ/ → [bugaŋbuga-ŋ] ‘eats fast’



Exponence Assignment via COPY. Once phonology receives the input, the copying operation assigns morphological affiliation to the newly created segments. The assumption is that copied material belongs to the reduplicative affix and not the morpheme it came from. This is how Consistency of Exponence is usually understood in phonology (McCarthy 1979, McCarthy and Prince 1993b): phonology does not assign morphological affiliation to epenthetic segments or reassign the affiliation of lexical segments, but it does supply segments to expone reduplicative morphemes. To continue with the Yidij, example, in /[]_{Ft} -buga-ŋ/, the reduplicated string copies from more than one morpheme, but there is no morpheme boundary inside the reduplicant. All copied segments are the reduplicant’s.

(17) Yidij exponence assignment via COPY: [bugaŋ_{Adv}-√buga-ŋ_{PAST}]



While this understanding of exponence borders on the obvious, it comes with interesting consequences

for morpheme-specific phonology.⁵ We adopt a diacritic theory of morpheme-specific rules (Pater 2008, a.o.), which allows constraints to track violations introduced by material from a specific morpheme or set of morphemes. This allows the diacritic markings for the base to diverge from those of the reduplicant (see Gouskova 2007 for a similar treatment within BRCT). We argue that Yidij morphemes fall into two broad classes with respect to the constraint FINALC, which triggers apocope in nominal bases but not reduplicative plurals (§4.4). FINALC also favors optional extra copying in verbal reduplication in §4.5.

3 Yidij Patterns

With this theoretical background in place, we now turn to Yidij. We start with an overview of aspects of Yidij phonology relevant to reduplication, and then break down the patterns based on analytic categories. First is total reduplication, the result of default copying of roots. Partial reduplication predominantly involves CV-final copying. We also discuss and analyze systematic exceptions.

3.1 Phonological Background

The consonant inventory of Yidij is given in Table 1. Only consonants other than stops and [w] can occur in coda position; nasal-stop clusters are generally homorganic⁶ and may be analyzed as prenasalized stops [mb, nd, nʃ, ŋg] (see fn.14). Nasal-stop sequences do not occur word-finally or utterance-initially.

	lab	alv	pal/retr	dor
stops	b	d	ʃ	g
nasals	m	n	ɲ	ŋ
liquids		l r	ɽ	
glides	w		y	

Table 1: Consonants of Yidij

Yidij contrasts three vowels, [i, u, a], which can be short or long. We chose to omit vowel length from transcription: first, vowels in reduplicants are always short, and second, vowel length is complex in ways that do not bear on our main points (see Dixon 1977a,b; Hayes 1999; Pruitt 2010).

According to Dixon, reduplicants form separate phonological words, with their own stresses (1977a p. 82). We assume that reduplicants are syllabic trochees, [σσ] (see Dixon 1977a,b; Hayes 1995). Some relevant phonotactics of Yidij are summarized in (19).

(19) Yidij phonotactics

- a. Roots and phonological words are generally disyllabic (there are a couple of exceptions in compounds, e.g. [já:-], as in [já:-wáŋgi] ‘upwards’ and [-(m)bar] in (b)). There are some prosodically non-cohering affixes, including *-ála*.

⁵This has an effect similar to Morphological Doubling Theory’s assumption that the base and the reduplicant are subject to potentially different co-phonologies, though there are differences in implementational detail (none of them will be consequential here, but see Pater 2006, 2008).

⁶But Dixon does not always transcribe them as such, possibly for reasons of orthographic convenience (thus, his <nɖ> may well stand for [nʃ]). Our transcriptions do not alter Dixon’s except for his symbol [ɖ], which we replace with [j] to avoid confusion with retroflex stops.

- b. Voiced stops are optionally prenasalized at a compound boundary (e.g., [baɾumbaɾ] ‘wattle tree grub’ vs. [baɾu-baɾ] in slow speech, cf. [maŋgumbaɾ], another grub species);
- c. The glide [y] does not occur after [i], although Dixon sometimes posits it underlyingly to explain morphological distributions (e.g., *galbiy-galbiy* ‘catfish PL’, pronounced [galbi]-[galbi]);
- d. There is a rule of [r, ɾ] deletion before [d], e.g., /RED-ɟabiɾ-da/ ɟabiɾ-ɟabi:da ‘on lots of flat rocks’.

3.2 Total Reduplication

Reduplication serves two productive functions in Yidiɲ: nominal reduplication marks plurals, and verbal reduplication is a manner modifier (“a lot”, “quickly”, etc.) Plural marking is apparently optional in Yidiɲ (Dixon 1977a:156). There are some “inherently reduplicated” nouns that do not have simplex counterparts; most of these appear to be diminutives (e.g., (20c)) or pluralia tantum. As shown in (20) and (21), disyllabic roots reduplicate in their entirety. The reduplicant may even have more material in it than the base, as in (21f), where /ɾ-d/ is simplified at the root-suffix boundary. All page numbers below are from Dixon 1977a; we include an exhaustive annotated list of all the reduplicative forms in his grammar as supplementary materials.

(20) Total σ CV reduplication

- | | | | | | |
|----|-------------|-------------------|-------|----------|--------|
| a. | ɟabu-ɟabu | ‘lots of ground’ | ɟabu | ‘ground’ | p. 157 |
| b. | buɟa-buɟa | ‘women’ | buɟa | ‘woman’ | 69 |
| c. | birbi-birbi | ‘small flat fish’ | (N/A) | | 478 |

(21) Total σ CVC reduplication

- | | | | | | |
|----|----------------|-------------------------|-------|-------------|-----------|
| a. | guran-guran | ‘tall (pl.)’ | guran | ‘tall’ | p. 533 |
| b. | ɲalal-ɲalal | ‘lots of big ones’ | ɲalal | ‘big’ | 69 |
| c. | buɟim-buɟim | ‘red bird species’ | N/A | (inherent) | 37 |
| d. | giram-giram | ‘cramp(s)’ | N/A | | 553 |
| e. | giram-giram-di | ‘with cramps’ | | | 336 |
| f. | ɟabiɾ-ɟabi-da | ‘on lots of flat rocks’ | ɟabiɾ | ‘flat rock’ | *[ɾd] 156 |

3.3 Partial reduplication

Partial reduplication in Yidiɲ includes several subpatterns. First are longer roots whose second vowel is followed by a consonant that cannot be a coda (see (22)). This includes obstruents and NC sequences, assuming those are single consonants.

(22) Partial σ CV reduplication: next C not a possible coda

- | | | | | | | |
|----|----------------|-------------------|-----------|-----------|-------------------|-------|
| a. | guda-gudaga | ‘dogs’ | gudaga | ‘dog’ | *g] _σ | p. 43 |
| b. | wagu-waguɟa | ‘men’ | waguɟa | ‘man’ | *ɟ] _σ | 69 |
| c. | maɟi-maɟinda-n | ‘keep walking up’ | maɟinda-n | ‘walk up’ | *nd] _σ | 233 |

Next are cases where there is an eligible coda to copy. These exhibit more complexity than the usual characterizations of Yidiɲ would suggest. For example, there are nominal roots followed by single-consonant suffixes, which are codas and which do not get copied. Some of these are derived by affixing a single-consonant suffix onto a reduplicated stem (e.g., *baɾa-baɾa-ŋ* has the -ŋ case suffix attach to *baɾa-baɾa*), but not all. All the examples, however, are consistent with root reduplication. Note, then, the

contrast between disyllabic bases that are morphologically complex (*guyga-guyga-m* in (23a)) and ones that consist of just a root *giram-giram* in (21d)).

(23) Eligible codas not copied when they are -C suffixes

a.	guyga-guyga-m	‘afternoon/evening’	498	cf. guyga-m	‘yesterday’
b.	gaʃi-gaʃi-n-bara	REDUP-long.way-GEN-ABS	144	gaʃi	‘long way’
c.	baʃa-baʃa-ŋ	‘fly species ERG’	126	cf. gala-mbaʃa	‘march fly’
d.	guʃu-guʃu-ŋ	‘rainbow ERG’	498	guʃu-guʃu	‘rainbow’
e.	gulu-gulu-y	‘black bream COMIT’	536	(N/A)	

There are also examples where there is an eligible coda, the root is longer than two syllables underlyingly, and reduplication follows the cross-linguistically common σ CV pattern (see (24)). Dixon cites one reduplicated example that has an invariant trisyllabic base, *mula-mulari* (others, such as *maʃinda-n* in (22c), are analytically ambiguous). The rest undergo vowel-zero alternations. While Dixon includes more examples of reduplication of this second type than the first, he is clear that there are more roots of the *mulari* type, and he analyzes final vowel deletion/apocope as a minor rule (see §4.4 for more).

(24) Partial σ CV reduplication: trisyllabic roots

	Pl	p.	ABS	ERG	gloss	
a.	/mulari/	mula-mulari	156	mulari	mulari-ŋgu	‘initiated man’
b.	/ʃambula/	ʃambu-ʃambul	69	ʃambul	ʃambula-ŋgu	‘two, a pair’
c.	/yabuʃu/	yabu-yabuʃ	69	yabuʃ	yabuʃu-ŋgu	‘girl, young woman’
d.	/ʃimuru/	ʃimu-ʃimur	156	ʃimur	ʃimuru-ŋgu	‘house’
e.		ʃimu-ʃimuru-la	533			‘houses-LOC’

Verbs are more complex still. They vary between σ CV and σ CVC patterns, optionally copying a coda consonant contributed by a single-C suffix (see (25)). Verbs can be derived from nominal plurals (see (25d)), in which case the full root is copied; the consonant /l/ cannot be left out. It is clear that whatever is happening in verbs, it transcends mere syllable copying (see §4.5 for full details).

(25) Total σ CVC reduplication: verbal examples

a.	buga(n)-buga-ŋ	‘eat fast (past)’	234	buga-n	‘eat!’
b.	bugaŋ-buga-ŋ	‘eat fast (present)’	234		
c.	wawa(-l)-wawa-l	‘look thoroughly’	234	cf. wawa	‘look/see!’
d.	[ŋalal-ŋalal] _{N-pl} -ala	‘got to be big (men)’	534	cf. ŋalal-ŋalal	‘lots of big ones’
	*ŋala-ŋalal-ala				

Some of the crucial evidence for syllable copying is supposed to come from roots longer than two syllables with a CVC second syllable. There are only three examples like this:

(26) Partial σ CVC reduplication

a.	gindal-gindalba	‘lizards’		gindalba	‘small lizard species’	553
b.	gunʃil-gunʃilbay	‘tiger snakes’		gunʃilbay	‘tiger snake’	554
				cf. gunʃi-n	‘to return’	554
c.	ʃugar-ʃugarba-n	‘to have an unsettled mind for a long time’		ʃugarba-n	‘to have an unsettled mind’	551

We found some evidence that at least (a) and (b) might be morphologically undersegmented. First,

there are several other names of reptile species that end in *-ba* and *-bay* (see (27)). We noted in (19a–b) that not all roots of Yidj are disyllabic; the *-ba(y)* strings could be parts of compounds in reptile names similar to *-bar* in names of grubs. If so, *guŋŋil-guŋŋil-bay* is just another example of root copying. As for (26c), its *-ba* could be one of two suffixes in Yidj, a locative and one meaning ‘one of a group’ (Dixon 1977a p. 560). See §4.5.2 for additional discussion of this example.

(27) Possible evidence for morphemic *-ba* and *-bay*

- a. guṭabay ‘small lizard sp.’
- b. gilbay ‘guana sp.’
- c. guṭbaba ‘lizard sp.’

Upon closer inspection, other examples of supposedly long stems showing partial reduplication turn out to be amenable to similar reanalyses. Take Dixon’s suggestion that *gala-galambaṛa* leaves off the first part of the nasal because homorganic nasal-stop clusters are not split by reduplication (Dixon 1977a p. 156). The second half of this word appears in another fly species name, and the nasal could be inserted by the rule discussed in (19b). In *wamba-wambaŋi* (see (28)), the nasal is a verbal conjugation marker. Since there are other explanations for these patterns, we conclude that the evidence for syllable copying is scant at best, and always amenable to alternative explanations.

(28) ND clusters

- | | | | | |
|----|--------------------|------------------|--------------------|-----------------------------|
| a. | gala-mbaṛa | gala-gala-mbaṛa | ‘march fly pl.’ | cf. baṛa-baṛa ‘fly species’ |
| | | | | cf. gala ‘a type of spear’ |
| b. | wamba-n ‘wait for’ | wamba-wamba-n-ŋi | ‘will have to wait | |
| | | redup-wait-lest | for a long time’ | p. 520 |

To summarize, the empirical picture in Yidj is consistent with the following generalizations:

- (29) Disyllabic roots are copied in their entirety;
- (30) Longer roots (including roots with vowel-zero alternations) are copied as σCV ;
- (31) Verbal reduplication copies σCV plus optionally single-consonant suffixes.

4 Yidj Analysis

The gist of our analysis⁷ is summarized in (32). We expand on these points in order in the remainder of this section.

(32) Analysis of Yidj in STS

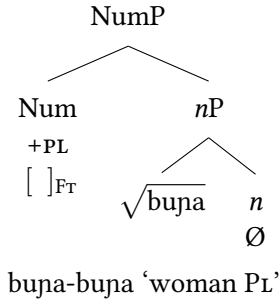
- a. The reduplicant is underlyingly a foot, []_{FT}.
- b. Roots that can fit into a foot are copied whole: *COPY-MORPH is low-ranked.
- c. Roots longer than two syllables reduplicate as σCV to fit the template, violating *COPY-SEG. NoCODA favors an open final syllable.
- d. Reduplication derivationally precedes apocope, driven by a lexically indexed FINALC_{+m}.
- e. Verbal stems are copied variably, $\sigma CV(-C)$, due to an interaction of FINALC_{+m} with copying constraints.

⁷Some parts of our analysis track that of Dyirbal in McCarthy et al. (2012), though the full pattern in Dyirbal is more complicated than this, with at least some trisyllabic roots reduplicating whole: *gulgiṛi-gulgiṛi* ‘lots of prettily painted men’ (Dixon 1972:242).

4.1 Nominal Morphology

Reduplication realizes a +PL feature in NumP, while verbal reduplication expresses some kind of an adverbial/manner adjunct (see §4.5).

(33) Structure for nominal/plural reduplication in Yidj



The phonological template for both plural reduplication and verbal manner reduplication is the same, an empty foot (see (34)). Yidj has other unproductive reduplicative affixes (e.g. diminutives) that are also foot-sized, and appear to be phonologically the same as the plural reduplicant. Since the nominal and verbal/adverbial reduplication differ phonologically in the extent of copying, we assume that one of them is indexed to FINALC_{+m}, a constraint that is independently motivated in the language (see §4.4ff.).

(34) Vocabulary insertion rules for reduplicative prefixes

- a. +PL ↔ []_{Ft}
- b. ADV ↔ []_{Ft+m} = indexed to FINALC_{+m}

4.2 Default: COPY-MORPH

We start with the analysis of (nominal) disyllabic roots. These are copied in their entirety (see (35)), which violates *COPY-MORPH. Copying anything less than a full root is disfavored by undominated FTBIN ("Feet must be binary at the syllabic level"). Disyllabic roots are also copied whole regardless of syllable shape: in (c-d), NoCODA favors partial copying (violating *COPY-SEG), so it must be outranked.

(35) Total copying of disyllabic stems

σCV:	[] _{Ft} + bupa 'woman'	FTBIN	*COPY-SEG	*COPY-MORPH	NoCODA
a.→	[σσ] _{Ft} bupa-bupa			*	
b.	[σ] _{Ft} bup-bupa	*W	*W	L	*W
σCVC:	[] _{Ft} + gu.ran 'tall'				
c.→	gu.ran-gu.ran			*	**
d.	gu.ra-gu.ran		*W	L	*L

This analysis extends straightforwardly to disyllabic roots that are followed by suffixes, which supply codas. In (36) is a minimal pair: a CV-final root and a CVC-final one, which get copied in their entirety. The form *guy.ga-guy.ga-m* 'in the evening' shows that the root alone is copied even when a coda is supplied by a suffix. The only way to copy a suffix in addition to the root in one derivational step is to deploy

COPY-SEG, which rules out candidate (36b). The suffix could be copied in a subsequent derivational step, but nothing favors that, so the derivation stops at (36a). A phonologically similar consonant-final stem *giram* will reduplicate whole (see (36c–d)).

(36) Suffixes not copied by reduplication

	[] _{Ft} + guy.ga-m 'in the evening'	FTBIN	*COPY-SEG	*COPY-MORPH	NoCODA
a. →	guy.ga-guy.ga-m			*	***
b.	guy.gam-guy.ga-m		*W	L	***W
	[] _{Ft} + gi.ram-ji 'with cramps'				
c. →	gi.ram-gi.ram-ji			*	**
d.	gi.ra-gi.ram-ji		*W	L	*L

Thus, the nasal is copied when it is in the same morpheme as the root.

4.3 Partial copying: COPY-SEG

Longer roots such as *mulari* cannot be reduplicated whole, since they do not fit into the disyllabic template (ruled out by undominated FTBIN). When the root is too long, copying must apply at the segmental level. Recall that *COPY-SEG is violated once per instance of copying, regardless of how many segments are copied. Between the three candidates in (37), markedness determines what to omit: NoCODA breaks the tie.

(37) Stopping short of a sonorant coda in trisyllabic roots

	[] _{Ft} + mu.la.ri _{rt} 'initiated man'	FTBIN	CODACOND	*COPY-SEG	*COPY-MORPH	NoCODA
a. →	[mu.la] _{Ft} - mu.la.ri _{rt}			*		
b.	[mu.lar] _{Ft} - mu.la.ri _{rt}			*		*W
c.	[mu.la.ri] _{Ft} - mu.la.ri _{rt}	*W		L	*W	

As we show in §4.4, this is also the first step in other reduplicative subpatterns, which involve further derivational steps that modify the base.

To sum up, so far, the basic pattern is that disyllabic roots are reduplicated whole regardless of shape, while longer roots are copied up to the second vowel. In our analysis, the default is to preferentially violate *COPY-MORPH, except where the template cannot accommodate the whole root—in which case NoCODA dictates how much to copy. Thus, syllable structure isn't irrelevant, but it is not what is being copied.

4.4 Opaque Interactions: An Argument for Serialism

We turn now to vowel-zero alternations and *rd*-cluster simplification. The main point of interest here is that these cases support a derivational analysis, which STS can provide without difficulty (unlike BRCT; see §5.1.4).

There are 98 roots in Dixon's corpus with vowel-zero alternations, such as [jimu-jimur] 'houses' (cf. [jimuru-la] 'house LOC') and /yabuɾu/ [yabuɾ] 'young girl' (recall (30)).⁸ According to Dixon, these redu-

⁸We follow Dixon in analyzing these alternations as deletion rather than epenthesis, but Dixon (1977a p. 36 and 62) gives some hints that this might not be entirely right (see also Hayes 1999). The quality of the 3rd (alternating) vowel is largely predictable either from the preceding or the following vowel, and there are even some stems that speakers produce with inconsistent quality,

plicate as vowel-final—which would be surprising if surface syllable structure were being copied. Derivationally, reduplication appears to precede apocope:⁹

- (38) An opaque/derivational account of [jimur-jimur] ‘houses’: copy first, then apocope

/[]_{Ft} - jimuru/ → [ji.mu]-[ji.mu]ru → [ji.mu]-[ji.mur]

We can achieve the same result in our framework: copying applies before apocope because the constraint that requires the template to be filled is ranked above the constraint that favors deletion in the base. There is an independent argument for having copying apply early: recall that reduplicative prefixes can retain segments that are lost in bases, e.g., /RED-jabiɾ-da/ → [jabiɾ-jabi_-da] ‘on lots of little rocks’. To account for the serial interaction between reduplication and consonant cluster simplification/apocope, we just need to ensure that constraints that favor the latter are dominated by HEADEDNESS[FT], which demands template satisfaction.

Let’s start with apocoping roots, shown in (40). This tableau shows that CV-final reduplicants win here for the same reason that they win in *mula-mulari*: NoCODA. Since Harmonic Serialism admits only one operation per derivational step, copying competes with apocope. To analyze apocope, we use a lexically indexed constraint FINALC_{+m}, defined in (39).¹⁰

- (39) FINALC_{+m}: Assign a violation for any vowel that is final in a phonological word. (Indexed to morphemes bearing +m diacritic)

- (40) Apocoping roots: reduplication step

	[] _{Ft} + jimuru _{+m}	HEAD[Ft]	FINALC _{+m}	*CPY-SEG	*CPY-MRPH	NoCODA	MaxV
a. →	[jimur] _{Ft} -jimuru _{+m}		*	*			
b.	[] _{Ft} -jimur _{+m}	*W	L	L		*W	*W
c.	[jimur] _{Ft} -jimuru _{+m}		*	*		*W	

The next step is to apocope. Here is where our assumptions about exponence assignment (§2.5) become relevant. The first string *jimu* is created by reduplicative copying and therefore does not bear the index +m, so its final vowel does not incur a violation of FINALC_{+m}.

- (41) Apocoping roots: deletion step

	[jimur] _{Ft} -jimuru _{+m}	HEAD[Ft]	FINALC _{+m}	*CPY-SEG	*CPY-MRPH	NoCODA	MaxV
a. →	[jimur] _{Ft} -jimur _{+m}					*	*
b.	[jimur] _{Ft} -jimuru _{+m}		*W			L	L

as in *wúbul* ~ *wubulu-ngu* (ERG) ~ *wubula-nda* (DAT) ‘lucky’. Speakers also reduce those vowels to schwa, which is something they do not do with non-alternating vowels: *gúbum* ~ *gúbumá-ni* ~ *gúbumǎ-ni* (GEN) ‘black pine’. We verified that there is a strong statistical tendency for vowels in the second and third syllable of such roots (Dixon lists all 98 of them on pp. 65-68) to be identical. The Observed/Expected ratios for matching vowel sequences in these roots are a-a: 1.719, i-i: 3.5, u-u: 1.693. For comparison, in 162 non-apocoping trisyllabic roots, the last two vowels’s O/E are a-a 0.953, i-i 1.342, and u-u 2.283. The full data files we used will be made available as supplementary materials. Despite this large degree of predictability, there are some stems in which the vowels are not copies: e.g., /jalamu/ ‘fresh, young’, /jurinu/ ‘leech’, /gambira/ ‘tablelands’, so we decided to go with deletion. Even if this is epenthesis, however, it only serves to reinforce our claim that there is no syllable copying. If the UR of [jimur-jimur] is /RED-jimur/, then the last consonant isn’t “underlyingly an onset” (contra Marantz 1982, Haugen and Hicks Kennard 2011, and McCarthy et al. 2012).

⁹Marantz (1982) is one of the few to attempt an analysis of this pattern. Since his theoretical assumptions do not allow copying/association to be interspersed with other rules, his account posits syllabification in the UR; this is at best incomplete as it would have to spell out a device (such as Morpheme Structure Constraints) for ensuring correct syllabification before phonology has applied.

¹⁰The name of the diacritic is a nod to Dixon’s characterization of final vowels in these stems as “morphophonemes” I, U, A.

In the case of *rd* clusters, the order of operations is similar: copy, then delete material in the base. The driver of deletion is *RD, a surface-true phonotactic constraint. The first step in deriving [ʃabiɫ-ʃabi-da] is copying a morpheme, violating *COPY-MORPH, rather than copying a substring of a morpheme, violating *COPY-SEG. In order for this to be optimal, HEAD[Ft] must outrank *RD—otherwise cluster simplification would overapply, *[ʃabi-ʃabi-da]. Copying less than a root, as in (42c), is correctly ruled out. In the next step, the base rhotic is deleted (see (43)), and the derivation converges: there is no reason to delete the rhotic from the reduplicative copy.

(42) Copying step for *rd*-simplification

	[] _{Ft} -ʃabiɫ-da	HEAD[Ft]	*RD	*CPY-SEG	*CPY-MRPH	MAX-C
a. →	[ʃabiɫ] _{Ft} -ʃabiɫ-da		*		*	
b.	[] _{Ft} ʃabi-da	*W	L		L	*W
c.	[ʃabi] _{Ft} -ʃabiɫ-da		*	*W	L	

(43) Cluster simplification step for *ʃabiɫ-ʃabi-da*

	[ʃabiɫ] _{Ft} -ʃabiɫ-da	HEAD[Ft]	*RD	*CPY-SEG	*CPY-MRPH	MAX-C
a. →	[ʃabiɫ] _{Ft} -ʃabi-da					*
b.	[ʃabiɫ] _{Ft} -ʃabiɫ-da		*W			L

The analysis of apocoping roots naturally groups them with the *mula-mulari* pattern; the only necessary assumption is that apocope happens after reduplication. In Yidij, apocope counterfeeds reduplicative copying: the final consonant of [jimur] surprisingly fails to be copied even though it is root-final on the surface. Cluster simplification, by contrast, counterbleeds reduplicative copying: the copying of the final consonant of [ʃabiɫ] is unexpected if all copying is surface-oriented. In our account, copying applies early, before phonological rules modify the base.¹¹ We argue in §5.1.4 that these facts are problematic for parallel BRCT.

4.5 Verbal reduplication

The last pattern we analyze is variable reduplication in verbs. As we have already established (recall (23)), single-consonant suffixes are not generally reduplicated in Yidij, even though they occupy the coda position. Verbal reduplication, however, can variably copy just the root, the root plus one of three conjugation markers (-*n*, -*l*, -*r*), or the root plus the tense or mood suffix (see (44)). The suffix copied into the reduplicant does not have to match the suffix that appears on the base, e.g., [bugan-buga-ɲ] ‘ate fast’. The URs assumed in (44) are elaborated in the following section; the [m] in *baygar-mbaygar* is likely the result of prenasalization at a compound boundary (recall (19)). We show vowel length here because it has a morphological function.

¹¹While we ignore length throughout, Dixon argues (1977a p. 69) that reduplication must precede lengthening, as well, since reduplicative vowels are always short (he assumes that vowel length is derived by a complicated set of rules that interact with stress).

(44) Verbal intensive reduplication: -C copying optional (pp. 211, 234 except where noted otherwise)

a.	/RED-wawa-l- μ /	wawal-wawa-:l	‘look thoroughly’
b.		wawa-wawa-:l	
c.	/RED-buga-n- η u/	bugan-buga- η	‘ate fast’
d.		buga-buga- η	‘ate fast’
e.	/RED-buga-n/	bugan-buga-n	‘eat fast!’
f.	/RED-buga- η /	buga η -buga- η	‘eats fast’
g.	/RED-ba η a- τ -[-anterior]/	ba η ar-ba η a-r	‘follow quickly!’
h.	/RED-ba η a- τ - μ /	ba η a τ -ba η a-: τ	‘followed quickly’
		ba η a-ba η a-: τ	p. 375
i.	/RED-bayga- τ /	bayga τ -mbayga- τ	‘feels very sore’

Dixon records 15 examples of C-final reduplicants, and 29 examples of V-final ones (of which many have a consonant in the right position that is ineligible for copying). As we demonstrate below, our analysis can accommodate these patterns with just a small modification. First, some background on the morphology of Yidj verbs.

4.5.1 The Morphology of verbs in Yidj

An abbreviated paradigm of verbal suffixes is shown in (45), the segmentations are ours rather than Dixon’s. Some of the suffixes undergo lexicalized apocope (recall §4.4): the past tense suffix (- η /- η u) shows synchronic alternations, and the present-future [- η] is a (possibly fossilized) reflex of /- η a/, which appears in various other verbal forms. As we explain below, conjugation markers are morphosyntactically empty morphemes, whose choice is determined lexically by the root much as in, say, Romance languages (see Oltra Massuet 1999). The imperative forms - n and - r in (45) could be derived from /- η a/ and / τ / via featural affixation of [-anterior]. There is some uncertainty about whether [- n] or [- η] is the conjugation marker; Dixon (1977a:211) assumes - n in part because of the reduplication patterns (given *bugan-buga: η*).

(45) Some verbal paradigms (Dixon 1977a p. 207)

imperative:	- n	- $\emptyset$	- r
present-future:	- η	- l	- τ
past	-: η /- η u	-: l	-: τ
comitative:	- η a- l	- l -ma η a- l	- τ -ma η a- l
‘coming’ aspect	- η a-da- n	-: l -da- n	-: τ -da- n
‘going’ aspect	- η a-li- n	-: l i- n	-: τ i- n
Number of verbs:	155	125	15

There is clear evidence of the morphological status of the - n and - l conjugation markers (see (46)–(47)). The affixes derive verbs from nouns, and - l is removed in imperatives. The - r conjugation includes only 15 roots, but there is some evidence that - r /- τ is an affix, as well (e.g., [dandadan] ‘(doctor) rub(s)’ vs. *danda- τ* ‘rub’). Dixon reports (p. 234) that all the examples he encountered reduplicate [r] and [τ], but he records *ba η a-ba η a: τ* on p. 375.

(46) Evidence for affixal status of - l

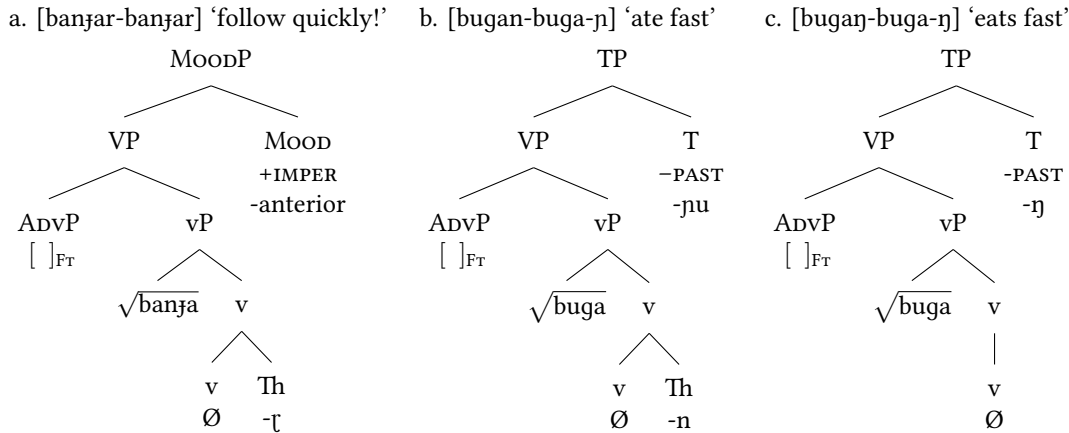
a.	wawa- l	‘to see, look’	wawa	‘look!’
b.	mala- l	‘to feel with the hand’	mala	‘palm of hand’

(47) Evidence for affixal status of *-n*

- a. bina-*n* ‘to hear, think’ bina ‘ear’
 b. gaɽba-*n* ‘to hide’ gaɽba ‘behind’
 c. gida-*n* ‘do quickly’ gida ‘quick’

We are now ready to consider the morphosyntactic structures. The interpretation of the verbal reduplicative prefix in Yidj is flexible/idiosyncratic in a way that is associated with structural proximity to the root (cf. Arad 2003, Marantz 2013, Kastner 2019, and many others). We assume the Adv modifier is attached low, as a sister to the root plus the verbalizing affix/theme.

(48) Structure for verbal manner reduplication



If these structures are correct, then at least some of the verbal reduplication patterns in Yidj constitute bracketing paradoxes: [bugaŋ-bugaŋ] and [bugan-bugaŋ] are unlikely to involve the copying of material at the same morphosyntactic height. As long as the entire string is available to phonology, however, the attested possibilities are not difficult to derive under our assumptions.

4.5.2 Analysis of variable affix copying

The main question is why verbal reduplication deviates from root-only copying: what compels the optional inclusion of *-C* suffixes? We suggest it is *FINALC_{+m}*, whose effects we extend here to other parts of the Yidj lexicon (see Hayes 1999). The constraint compels lexicalized apocope in nominal roots (/ʃimuru_{+m}/ ‘house’ reduplicating as *ʃimu-ʃimur*, §4.4) and in various suffixes, such as /-ŋu_{+m}/ ‘+PAST’. *FINALC* also affects the adverbial reduplicative prefix, variably forcing an extra consonant into the template to relieve vowel-finality. Copying targets either just the root or a root plus the following consonant from a suffix.

Since verbal and nominal reduplication differ, we posit distinct allomorphs for the reduplicants: both are empty feet, but the verbal affix alone is indexed to *FINALC_{+m}* (recall (34)). The variation between copying just the root (as in (49)) and the root plus one extra consonant (as in (50)) arises from varying the ranking of *FINALC_{+m}* and **COPY-SEG*.¹² Based on nominal reduplication patterns, the two constraints do not crucially conflict, since *FINALC_{+m}* is irrelevant to the nominal reduplicant. In verbal reduplication, *FINALC_{+m}* is relevant, so its reranking with **COPY-SEG* has consequences.

We start by deriving the vowel-final variant, which is favored if **COPY-SEG* dominates *FINALC_{+m}*: a root alone is copied, and the preference for a consonant-final reduplicative morpheme is not satisfied. The

¹² A reviewer asks what the nature of the variable ranking is and what the predicted rates of variation are. The information available on the language does not allow for a more detailed analysis of rates. As for the nature of the variable ranking, pretty much any theory of variation in a constraint-based framework will do the job, be it MaxEnt or Noisy HG or Stochastic OT (see Coetzee and Pater 2011 for review). Our analysis does not adjudicate between theories of variation in any way.

derivation in (49) shows that the root is copied to fill out the headless foot, and then the sequence /-n-ju/ is gradually reduced to [-n], since the -ju suffix is also subject to the lexicalized FINALC_{+m} requirement. This pattern is indistinguishable from nominal reduplication of underlyingly vowel-final roots such as [jimu-jimur] ‘houses’.

(49) Variable reduplication in verbs: deriving [buga-bugan]

Step 1: copy root

	/Ft[] _{+m} -buga-n-ju _{+m} /	Hd[Ft]	*n-j	*CPY-SEG	FINC _{+m}	*CPY-M	MAXV	MAXC	NoCod
a. →	Ft[buga] _{+m} -buga-n-ju _{+m}				**	*			*
b.	Ft[] _{+m} -buga-n-ju _{+m}	*W			*L	L			*
c.	Ft[bugan] _{+m} -buga-n-ju _{+m}			*W	*L	L			**W

Step 2: simplify cluster

	Ft[buga] _{+m} -buga-n-ju _{+m}	Hd[Ft]	*n-j	*CPY-SEG	FINC _{+m}	*CPY-M	MAXV	MAXC	NoCod
d. →	Ft[buga] _{+m} -buga-ju _{+m}				**			*	
e.	Ft[buga] _{+m} -buga-n-ju _{+m}		*W		**			L	*W

Step 3: apocope

	Ft[buga] _{+m} -buga-ju _{+m}	Hd[Ft]	*n-j	*CPY-SEG	FINC _{+m}	*CPY-M	MAX-V	MAXC	NoCod
f. →	Ft[buga] _{+m} -buga-j _{+m}						*		*
g.	Ft[buga] _{+m} -buga-ju _{+m}				*W		L		L

In (50) we derive the consonant-final variant. This arises if FINALC_{+m} is ranked above *COPY-SEG: it forces the copying of a segmental string that extends beyond the root to the next consonant. Filling out the template must happen early in the derivation, and the choices are to copy at the segmental level or the morpheme level; FINALC_{+m} favors the otherwise dispreferred option of segmental copying. The rest of the derivation proceeds through the steps of simplifying the cluster and apocope:

(50) Variable reduplication in verbs: deriving [bugan-bugan]

Step 1: copy segmental string, root+C affix

	/Ft[] _{+m} -buga-n-ju _{+m} /	Hd[Ft]	FINC _{+m}	*CPY-SEG	*CPY-MRPH	MAXV	MAXC	NoCod
a. →	Ft[bugan] _{+m} -buga-n-ju _{+m}		*	*				**
b.	Ft[] _{+m} -buga-n-ju _{+m}	*W	*	L				*L
c.	Ft[buga] _{+m} -buga-n-ju _{+m}		**W	L	*W			*L

Step 2: simplify cluster

	Ft[bugan] _{+m} -buga-n-nu _{+m}	HD[Ft]	*n-ŋ	FINC _{+m}	*CPY-SEG	*CPY-MRPH	MAXV	MAXC	NoCod
d.→	Ft[bugan] _{+m} -buga-nu _{+m}			*				*	
e.	Ft[bugan] _{+m} - buga-n-nu _{+m}		*W	*				L	*W

Step 3: apocope

	Ft[bugan] _{+m} -buga-n-nu _{+m}	HD[Ft]	*n-ŋ	FINC _{+m}	*CPY-SEG	*CPY-MRPH	MAX-V	MAXC	NoCod
f.→	Ft[bugan] _{+m} -buga-n _{+m}						*		*
g.	Ft[bugan] _{+m} -buga-nu _{+m}			*W			L		L

The only other pattern in (44) left to explain is the alternation between rhotics, which we attribute to a floating feature.¹³ The available evidence suggests that when the rhotic is copied, it always matches the rhotic in the base (unlike nasals, where Dixon records mismatched place in [bugan-bugan]). As long as the floating feature is docked on the first derivational step—which will follow if *FLOAT (Wolf 2007) is undominated—then the rhotic will take along its [anterior] value if it is copied, /_{Ft}[]_{+m}-banʃar-[-anterior]/ → _{Ft}[]_{+m}-banʃar → banʃar-banʃar. This follows given our assumptions about segmental copying: it cannot leave behind a feature.

This analysis predicts that verbal roots longer than two syllables should variably copy σ CVC, since the ranking will favor a consonant-final copy regardless of whether that last consonant belongs to a -C morpheme or is part of the root. The prediction is not easy to test since most verbal roots (95%) are disyllabic. Of the few trisyllabic roots, only two are given in reduplicated contexts (see (51a–b)). These forms are consistent with the prediction, but there are alternative explanations for them (it is possible that [-ba] is a suffix in [ʃugarban], and the [nd] cluster in [maʃindan] is indivisible because NC sequences generally are indivisible in the language).¹⁴ Of the remaining trisyllabic roots (see (51)), some couldn't copy the first consonant from the third syllable for phonotactic reasons (51g–m), and several appear to be morphologically complex. Either way, we predict that verbal reduplicants should variably end in a consonant.

¹³The alternative is to assume that imperative [-r] and past tense [-ɾ] are unrelated allomorphs, in which case the account of reduplication is trivial: whichever affix appears, it is always copied.

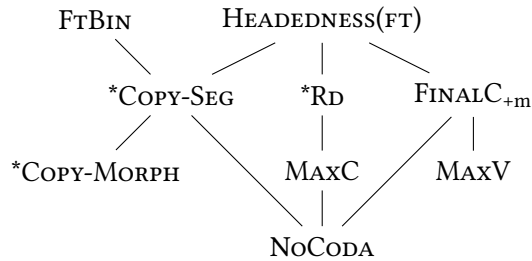
¹⁴To investigate the status of NC clusters, we submitted Dixon's lexicon of Yidiñ to the CompSeg learner (Gouskova and Stanton 2021). The learner analyzes the statistical distribution of consonant sequences, and posits complex segment representations for sequences that are more likely to occur together than apart. Our corpus was 1,000 items—Dixon's dictionary conservatively divided into stems and affixes. The results support the treatment of homorganic nasal-stop clusters as prenasalized stops: [mb, nd, nʃ, ŋg]. Thus, we have quantitative/learning-theoretic evidence for Nash's (1979) suggestion that NC sequences are prenasalized stops. In our theory, these complex segments can only be copied whole or not at all, and in all of Dixon's examples, they are left behind when the alternative is to put them in a position where they are phonotactically banned (word/foot-finally). See https://compseg.lingexp.org/demo_sim_detail/yidiny.

(51) Dixon’s trisyllabic verb examples (p. 208)

			Redup. reported?	Our notes
a.	jugarba-n	‘have unsettled mind’	jugar-jugarba-n	
b.	maʒinda-n	‘walk up’	maji-maʒinda-n	*maʒind] _{wd}
c.	bambaɽa-n	‘be frightened, nervous’	no	
d.	daɲʒiri-n	‘feel frisky’		
e.	ʃaʒama-n	‘jump over’		
f.	yilari-n	‘be scattered, spread about’		
g.	binarɽa-l	‘tell, warn’	no	cf. bina-n ‘hear’
h.	waymbala-n	‘roll’		cf. waymba-n ‘fly’
i.	barganda-n	‘pass by’		*bargand] _{wd}
j.	dandada-n	‘(doctor) rub(s)’		*dandad] _{wd}
				cf. danda-ɽ ‘rub’
k.	dandaba-n	‘dance around, feeling lively’		*dandab] _{wd} , -ba?
l.	daraba-n	‘shake, rinse mouth, etc.’		*darab] _{wd}
m.	wuɲaba-n	‘look for [meat], hunt’		*wuɲab] _{wd}
				cf. wuɲa-n ‘drink, swallow’

This concludes our analysis of Yidj̄n reduplication. The main rankings we argued for are in (52). FINALC_{+m} and *COPY-SEG are in a variable ranking (not shown).

(52) The rankings needed for Yidj̄n reduplication



5 Alternatives

We now compare our theory with alternatives, focusing on the two most popular ones: Base-Reduplicant Correspondence Theory (BRCT, McCarthy and Prince, 1994b,a, 1995 et seq.) and Morphological Doubling Theory (MDT, Inkelas and Zoll 2005 et seq.). We start by showing how BRCT derives templatic generalizations, since MDT’s approach builds on similar ideas. We then consider the specific challenges that Yidj̄n presents for these approaches. BRCT can mimic our analysis of conditional reduplication of disyllabic vs. longer roots, but fails at opacity effects. MDT, on the other hand, cannot derive even the basic pattern of C/V-finality, and it struggles with bracketing paradoxes, both in Yidj̄n and elsewhere.

5.1 Base-Reduplicant Correspondence Theory

5.1.1 Basic Correspondence Theory

McCarthy & Prince’s (1994b; 1994a; 1995) work on reduplication has been influential for several reasons, perhaps the main one being that it developed a theory of faithfulness constraints that worked uniformly

for input-output mappings and base-reduplicant (mis-)matches (as well as output-output correspondence Kenstowicz, 1996; Benua, 1997, etc.). In Correspondence Theory, the reduplicant is an abstract morpheme RED, which invokes its own correspondence relation $\mathfrak{R}$ with the base (see (53)). This correspondence relation is tracked by special base-reduplicant versions of MAX, DEP, IDENT, ANCHOR, and so on. These same faithfulness constraints also legislate input-output mismatches. Thus, in [mula-mulari], the base is completely faithful to the input /mulari/, satisfying MAX-IO. But the [ri] portion of the output/base lacks a correspondent in the reduplicative substring, which would incur two MAX-BR violations:

- (53) /RED-m₁u₂l₃a₄r₅i₆/ *Input-Output* correspondences: {m₁:m₁₁, u₂:u₁₂, l₃:l₁₃, a₄:a₁₄, r₅:r₁₅, i₆:i₁₆}
 [m₇u₈l₉a₁₀-m₁₁u₁₂l₁₃a₁₄r₁₅i₁₆] *Base-Reduplicant* correspondences: {m₇:m₁₁, u₈:u₁₂, l₉:l₁₃, a₁₀:a₁₄,
 Ø:r₁₅, Ø:i₁₆}

The argument for the faithfulness constraints being the same in base-reduplicant and input-output correspondence rests on a collection of cases that are argued to motivate various identity requirements in the two domains.

5.1.2 The Emergence of the Unmarked and Generalized Template Theory

Faithfulness demands identity, which partial reduplication fails to achieve. In OT, whenever faithfulness is violated, markedness is to blame. The answer to which markedness constraints drive partial reduplication has varied throughout the history of OT, so we cannot speak of a single parallel OT approach to reduplication or templates. Thus, McCarthy and Prince (1993b) use templatic constraints such as RED=FT. These were quickly abandoned¹⁵ in favor of deriving some aspects of templatic shape from The Emergence of the Unmarked, and others from Generalized Template Theory. The Emergence of the Unmarked (TETU, McCarthy and Prince 1994a) formalizes the intuition that the reduplicant copies only the unmarked material from the base because certain markedness constraints are inviolable within reduplicants—for example, syllables must be codaless and have onsets (cf. Steriade 1988 for a precursor). Generalized Template Theory develops the idea that reduplicative morphemes may be required to satisfy minimal size requirements or conversely be limited in size because they are morphologically roots or affixes (McCarthy and Prince 1994b; Urbanczyk 2006, a.o.). Work on atemplatic reduplication (e.g., Gafos 1998) is within the spirit of Generalized Template Theory in that it abandons templates.

To see why the usual characterization of Yidiñ is problematic for this approach, consider McCarthy & Prince's analysis of Diyari (Pama-Nyungan, Austin 1981), where reduplication also produces disyllabic phonological words. McCarthy & Prince's intuitive take on Diyari is that it reduplicates the language's unmarked phonological word. In non-reduplicated words of Diyari, the constraints that favor such a word are violable, and therefore outranked by input-output faithfulness. In reduplicants, however, the markedness constraints get full reign, favoring only perfectly aligned disyllabic monopod Pwds. The classic TETU ranking is FAITH-IO»MARKEDNESS»FAITH-BR:

- (54) Diyari reduplication (Austin 1981p. 65, McCarthy and Prince 1994a)
 a. [(wiḷa)_{FT}(pina)]_{Pwd} 'old woman' [(wiḷa)]-[(wiḷa)(pina)] 'little old woman'
 b. [(kinṯa)_{FT}la]_{Pwd} 'dog' [(kinṯa)]-[(kinṯa)la] 'little dog'

McCarthy & Prince's TETU/BRCT analysis:

MAX-IO » FTBIN » PARSE-σ » ALL-FT-L » ALL-FT-R » MAX-BR

¹⁵The other reason to abandon templatic constraints is the Kager-Hamilton Conundrum: the prediction that templatic size restrictions should be backcopied onto the base if both MAX-BR and RED=FT dominate MAX-IO, so that /RED-mulari/ → [mula-mula] while /mulari/ → *mulari* (Inkelas and Zoll 2005; Caballero 2006; Gouskova 2007).

In Diyari, the reduplicative prosodic word must be vowel-final, just like words in general must be vowel-final. In Yidj, by contrast, prosodic words may be consonant-final. The problem with Yidj (as typically described) is that if consonants are allowed in reduplicant position at all, MAX-BR will always favor their copying, as in *[mu.lar-mu.la.ri]. MAX-BR would always rather have a bigger reduplicant. This is the same problem that Copy-and-Association theory faced with Yidj (recall §1): since the reduplicant can be maximally CVCCVC, why does it not always take up the maximal expansion of the template?

To explain why a consonant is only copied when it is a coda in the base, McCarthy and Prince (1993b) have to invoke a constraint for cases like Yidj, which they dub STROLE. The constraint demands that segments in correspondence have the same structural (prosodic) roles. STROLE is not formalized, but in other prosodic morphology work, the complexities of defining such a relation become apparent (see in particular Ito et al. 1996, sec. 5 and McCarthy 2000). To analyze a Japanese secret language with foot reversal (e.g., [karaoke] → [okekara]), Ito et al. (1996) define constraints on each part of the prosodic constituent, so a foot equivalent of STROLE is something like six separate constraints. Zooming out from these specifics, however, the problem with STROLE is that it is a reduplication-specific faithfulness constraint with no analog in the input-output domain (McCarthy et al. 2012). No known languages contrast syllabification, much as no known languages contrast foot boundary positions (though obviously, plenty of languages have stress contrasts; see McCarthy and Pruitt 2013 for how to approach this within HS). Interestingly, the only known cases that motivate prosodic faithfulness of the sort Ito et al. (1996) develop for the Japanese secret language are other secret languages, never morphological rules in natural languages (see Moreton et al. 2020 and §5.4.4).

As we will see in §5.2, MDT’s approach to templatic generalizations is TETU in a cophonology implementation. Before looking at MDT, let us consider how BRCT handles Yidj in our reframing.

5.1.3 Root Copying via Anchoring

Recall our generalization: disyllabic roots are copied whole whether they end in consonants or vowels, while longer roots are cut off after the second vowel. In BRCT, this can be attributed to a violable constraint that demands the rightmost segment of the reduplicant to have a correspondent at the right edge of a root, ANCHOR-R-Rt-BR. This is a positional faithfulness version of McCarthy & Prince’s (1995) ANCHOR:

- (55) ANCHOR-R-Rt-BR: Assign a violation mark for every $S1\Re S2$ if $S1$ is at the right edge of RED and $S2$ is not at the right edge of a Root.

Ranking ANCHOR-R-Rt above NoCODA, as in (56), derives the contrast between *guyga-guyga-m* and *giram-giram*. (PWD=FT is a cover constraint for the Diyari-style TETU ranking). In this same ranking, *mulari* is copied as *mula-*: the constraints enforcing disyllabicity of RED (PWD=FT) dominate ANCHOR. ANCHOR is violated in both *mula-mulari* and *mular-mulari*, and NoCODA prefers *mula-*.

- (56) BRCT account: positional faithfulness to root-final consonants in reduplication

		MAX-IO	PWD=FT	ANCHOR-R-Rt-BR	NoCODA	MAX-BR
/RED- $\sqrt{\text{guyga-m}}$ /	a. →guyga ₁ -guyga ₁ -m				***	*
	b. guyga ₁ m-guyga ₁ -m			*W	****W	L
/RED- $\sqrt{\text{giram}}$ /	c. →giram ₁ -giram ₁				**	
	d. gira-giram ₁			*W	*L	*W
	e. gira-gira	*W			L	
/RED- $\sqrt{\text{mulari}}$ /	f. →mula-mulari ₁		*	*		**
	g. mular-mulari ₁		*	*	*W	*L
	h. mulari ₁ -mulari ₁		**W	L		L
	i. mula-mula	**W	L	L		L

As we show in §5.1.4, however, ANCHOR does not handle the root-final segments correctly in other contexts.

5.1.4 Problems for BRCT Account

When the focus is expanded to apocope and RD cluster simplification (§4.4), BRCT struggles. It can explain why the final vowel deletes in /jimuru/ [jimur] ‘house’ using the same mechanism as our own, indexed FINALC. But it is a mystery why the [r] is not reduplicated (see (57b)). The ranking favors greater faithfulness to the right edge, packing the reduplicative foot to the max. The opposite problem arises with RD clusters: the option of omitting [ɾ] from the reduplicant results in a better resemblance between reduplicant and base (see (57f)). This latter pattern requires input-reduplicant faithfulness in McCarthy & Prince’s (McCarthy and Prince 1995) original theory, which, as they point out themselves, adds enough power to destroy various predictions about the distribution of marked structure in reduplication. And, of course, favoring faithfulness to the input material in this case will make it even harder to explain why [jimu-jimur] succeeds.

(57) Problems for a Correspondence Account: (b) and (f) beat the actual winners

	/RED-jimuru _{+m} /	*RD	FINALC _{+m}	IO-MAX	PWD=FT	BR-ANCH-R(RT)	NoCod	BR-MAX
a.	→ jimu-jimur _{+m}			*		*	*	*
b.	jimur-jimur _{+m}			*		L	**W	L
c.	jimu-jimuru _{+m}		*W(u _{+m})	L	*W	*	L	**W
	/RED-jabiɾ-da/							
d.	→ jabiɾ-jabi-da			*		*	*	
e.	jabiɾ-jabiɾ-da	*W		L	*W	L	**W	
f.	jabi-jabi-da			*		L	L	

If our characterization of Yidj is wrong and the Dixonian generalizations are right—that is, if Yidj does copy syllables—then these patterns are again unexplained. Syllable copying satisfies a high-ranking STROLE (see §5.1.2), but it is violated in *jimu-jimur* and *jabiɾ-jabi-da*—and it is unclear why. Both (57b) and (f) satisfy STROLE better than the winners.¹⁶

The indivisibility of NC clusters is also a mystery for BRCT. For us, NC clusters are not split by copying because they are single segments, which COPY-SEG cannot split. Nothing favors copying more than the template requires. In BRCT, however, nasals should be copied into the reduplicant whenever phonotactics permit because copying more satisfies faithfulness better. Presumably something prevents a nasal from standing in correspondence with a prenasalized stop, but again, it is unclear what that might be. Yidj readily tolerates mismatches in nasality between reduplicative copies and the base in forms like [baygaɾ-mbaygaɾ] ‘feeling very sore’.

¹⁶ A reviewer suggests that certain extensions to BRCT could circumvent these problems: BRCT combined with Stratal OT (Bermúdez-Otero 2018; Jaker and Kiparsky 2020), or BRCT combined with OO-Faith (Kenstowicz 1996; Benua 1997, a.o.). For a Stratal OT account of [jimu-jimur] to work, the apocope stratum has to follow the reduplication stratum: the form that reduplication applies to must have a third vowel so that the /r/ does not get copied. But this is backwards from how apocope works. It is a lexicalized rule in Yidj, and as such it is supposed to apply early—and if it does, then reduplication should copy the [r]. Of course, MDT (discussed in §5.2) is often characterized as an improved descendant of Stratal OT, and we demonstrate that it cannot handle even the basic reduplication pattern, let alone the edge cases. As for OO-Faith, it also predicts the exact wrong outcome for apocope cases. The usual assumption is that derived forms should be faithful to some basic morphological form from which it is derived. For the plural ‘houses’, that should be ‘house’—not its ergative or another case-affixed form. So it is really unclear what aspect of the reduplication in [jimu-jimur] could be explained away by appealing to faithfulness to [jimur]. If anything, the reduplicated form supplies evidence that the word is derived from a UR that is preserved better in other derived forms, not in the form that is the most likely candidate for the derivational base.

Thus, we add to McCarthy et al.'s (2012) arguments against BRCT in their original presentation of STS. They mainly criticize the theory for overprediction, since it can easily derive two arguably unattested patterns: *coda skipping*, where /CVCCVC. . ./ is reduplicated as CVCV, and what we call *foot packing*, which copies an extra C to satisfy MAX-BR better. McCarthy et al.'s typological survey of disyllabic reduplication finds only σ CV patterns with contiguous copying; medial codas are not skipped and feet are not packed. In the few cases of non-contiguous reduplication, such as Seediq and Sanskrit, deletion is independently motivated elsewhere in the language. None of the cases look like (58).

(58) Predictions of BRCT analysis of disyllabic reduplication

		Ranking in BRCT
a.	<i>Coda skipping</i> : ki_ta-kintala	MAX-IO»NoCODA, FtBIN»MAX-BR, CONTIG
b.	<i>Foot packing</i> : kintal_kintala	MAX-IO» FtBIN»MAX-BR»NoCODA

STS avoids both predictions. First, COPY-SEG cannot produce discontinuous reduplicants without extra deletion steps, which must be independently motivated (McCarthy et al. 2012, §6.6). Second, since there is no faithfulness pressure to maximally copy from the base, nothing favors including an extra coda into the foot-sized reduplicant (unless there is an independent reason, e.g., to satisfy FINALC).

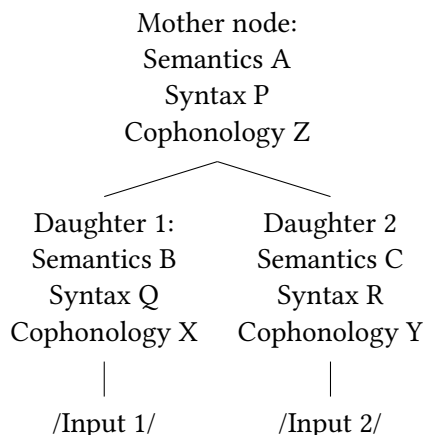
There has been a shift in phonology away from attempting to derive a restrictive typology through restricting the theory, and toward learnability-based explanations for typological gaps (see §5.4). Thus, not everyone considers overprediction to be a problem. But it should be considered fatal to a theory if it cannot account for an attested pattern. We argue that BRCT cannot gracefully handle the apocope and *rd*-deletion patterns in Yidj, largely because the reason for these patterns is their derivational history. Harmonic Serialism is designed to harness derivational history in its very architecture, and therefore opaque patterns are not difficult; they are indeed predicted.

5.2 Morphological Doubling Theory

5.2.1 Basic templatic generalizations of Yidj cannot be derived in MDT

Morphological Doubling Theory (Inkelas and Zoll 2005) espouses a sign-based view of morphology: syntactic, semantic, morphological, and phonological levels of representation are coextensive (see 59). In MDT, reduplication entails copying a morphosyntactic subtree, which is then associated to its own cophonology. Copying logically precedes any phonological modifications:

(59) A sign-based theory of morphology (Inkelas and Zoll 2005 pp. 76, 88)



For Yidj nominal reduplication, the semantics of the reduplicative daughter would specify “plural”, and its phonology would legislate templatic generalizations such as disyllabicity. MDT’s take on templates is very similar to BRCT’s TETU (§5.1.2), only instead of a single FAITH-BR≫MARKEDNESS≫FAITH-IO ranking, it is split into two cophonologies (see (60)). There is no correspondence relation between bases and reduplicants; hence only one type of faithfulness: “In cophonology theory phonological rules or constraints are never themselves indexed to particular morphological contexts; it is the entire bundle of rules or the specific ranking of constraints constituting the cophonology itself which is morphologically indexed” (Inkelas and Zoll 2005 p. 71).

(60) Diyari and Yidj reduplicant disyllabicity in MDT with Cophonologies

- a. Base cophonology: FAITH≫FtBIN≫PARSE-σ≫ALL-Ft-L≫ALL-Ft-R
- b. Reduplicant cophonology: FtBIN≫PARSE-σ≫ALL-Ft-L≫ALL-Ft-R≫FAITH

In terms of typological predictions, MDT shares some liabilities with BRCT. For example, it predicts coda-skipping: /Pl-pintuk/ will reduplicate as *pitu-pintuk* if the reduplicant’s cophonology ranks NoCODA over MAX and CONTIG.

When it comes to Yidj, MDT struggles with the very basics. In Yidj reduplication, consonant-finality is predictable from the shape of the root. In the traditional framing, the choice depends on syllabification (see Haugen and Hicks Kennard (2011) on this point for MDT). In our reframing, codas are allowed only when the reduplicant copies a whole root, and variably in verbal reduplication. We saw that BRCT could achieve some subtlety in deriving conditional coda copying when the root ended in a consonant via ANCHOR-Rt. But in MDT this move is not available, because the theory does not allow constraint indexation to roots. No consistent cophonology would truncate *mulari* at [a] without also truncating consonant-final disyllabic roots in the same way. This is spelled out in (61a–b):

(61) MDT: no consistent ranking for truncation in longer stems vs. disyllabic ones

		WD=Ft	MAX	NoCODA	FINALC
a.	/giram/ giram > gira		W	L	W
b.	/mulari/ mula > mular		L	W	L
c.	/jimuru/ jimu > jimur		L	W	L
d.	/jabit̪/ jabit̪ > jabi		W	L	W

Beyond basics, MDT also sheds no light on apocopating roots (/Pl-jimuru/→[jimu-jimur]). Cophonology Theory approaches morpheme-specific rules via indexing the morphemes to a separate cophonology: perhaps something like FINALC≫MAX for apocopating roots while MAX≫FINALC for regular ones. But FINALC cannot be ranked in a consistent spot for all reduplicative affixes (see (61c–d)). The only solution is to specify two cophonologies for reduplicative plurals: one in which FINALC dominates MAX, and another in which it is dominated by MAX. And it so happens that all the roots whose reduplicative affixes are specified for MAX≫FINALC happen to be /σCVCV/ underlyingly. The shape of reduplicative affixes in Yidj is predictable from the morpho-phonology of the roots, but there is no way to capture this in Cophonology Theory.

5.2.2 Bracketing Paradoxes in a Sign-Based Theory: Kihehe

As a sign-based theory of morphology, MDT runs into higher-level problems, as well: because it assumes complete homology between syntactic, semantic, morphological, and phonological domains, it cannot gracefully account for bracketing paradoxes. We saw such a paradox in Yidj verbal reduplication (§4.5): the copied strings do not form consistent morphosyntactic constituents. There, the situation was complicated by variation, so let’s consider a cleaner example from Kihehe (Odden and Odden 1985, Marantz 1987,

Nyamahanga 2025). Kihehe moderative reduplication (“a bit”) copies a consonant-initial root and any following suffixes (e.g., *-ol*, *-a*, *-e*), but not prefixes (see (62a–d)). If the root is vowel-initial, as in (62e–g), the preceding prefix is copied:

- (62) Kihehe reduplication: extent of copying determined by phonological shape of root (Marantz 1987:203, Odden and Odden 1985)

consonant-initial stems: prefixes not copied

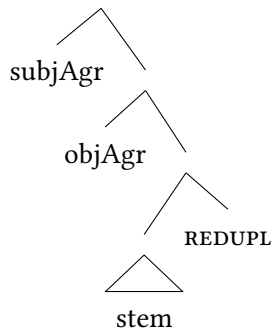
- | | | | | |
|----|---------------|----------------------|------------------------|-----------------|
| a. | kú-tov-a | kú-tova-tóva | ‘to beat a bit’ | kú- ‘inf’ |
| b. | kú-gohom-ól-a | kú-gohomola-gohomóla | ‘to cough a bit’ | |
| c. | tu-gul-ít-e | tu-gulite-gulíte | ‘we shopped a bit’ | tu- ‘we’ |
| d. | kú-fi-gul-a | kú-fi-gula-gúla | ‘to buy a bit of them’ | fi- Agr. prefix |

vowel-initial stems: prefix copied

- | | | | | |
|----|----------------|----------------------|----------------------------|-------------------------|
| e. | kw-iit-a | kwíita-kwíita | ‘to pour a bit’ | kw- ‘inf’ |
| f. | kú-lw-iit-a | kú-lwíita-lwíita | ‘to pour it a bit’ | lw- ‘it’ |
| g. | va-gy-aan-ik-a | va-gyaanika-gyaaníka | ‘they will dry them a bit’ | va-, gy-: Agr. prefixes |

The basic pattern, shown by the consonant-initial roots, respects the morphosyntactic position of the reduplicative affix. As an adverbial, it takes scope over the stem but not the subject/object prefixes; indeed the prefixes’ systematic exclusion from reduplication of consonant-initial stems shows that their position is lower. In (63) is the structure, and (64) shows the vocabulary items:

- (63) Morphological structure of Kihehe moderative verbs, following Marantz (1987)



- (64) Vocabulary Items in Kihehe [kw-íita-kwíita] ‘to pour a bit’

- | | | | |
|----|----------------------------|----|-----------|
| a. | [DEG] ↔ [] _{Pwd} | c. | INF ↔ ku- |
| b. | √ _{pour} ↔ iit- | d. | ASP ↔ -a |
| | √ _{beat} ↔ tov- | | |

For consonant-final roots, the default pattern is to copy the XP that is the sister¹⁷ to the reduplicative suffix, filling out the empty prosodic word template.

- (65) Kihehe consonant-initial stems (default): copy XP sister

/ku-tov-a-[] _{pwd} /	ONS	HD(Pwd)	*COPY-SEG	ID[syll]	*COPY-XP
a. → ku-tov-a-[tova] _{pwd}					*
b. ku-tov-a-[] _{pwd}		*W			L
c. ku-tov-a-[kutova] _{pwd}			*W		L

¹⁷We assume that, just like COPY-MORPH, the operation COPY-XP places the copied constituent next to the original—thus, it is impossible for [kutova] to be a copied XP, it can only be a copied string.

Vowel-initial stems deviate from the base pattern, expanding the reduplicated string to include one (complex) consonant to the left of the stem. The prefix must undergo gliding first to satisfy *ONSET* (/u/→[w], violating *IDENT*[syllabic]), so at Step 1, the empty prosodic word is tolerated. The empty word is filled out at step 2, via segmental string copying:

(66) Kihehe vowel-initial stems

Step 1: fix hiatus

/ku-iit-a-[] _{pwd} /	ONS	HD(Pwd)	*COPY-SEG	Id[syll]	*COPY-XP
a.→ kw-iit-a-[] _{pwd}		*		*	
b. ku-iit-a-[iita] _{pwd}	*W	L		L	*W
c. ku-iit-a-[ku-iit-a] _{pwd}	**W	L		L	*W
d. ku-iit-a-[] _{pwd}	*W	*		L	

Step 2: copy string, satisfying *ONSET*

kw-iit-a-[] _{pwd}	ONS	HD(Pwd)	*COPY-SEG	Id[syll]	*COPY-XP
a.→ kw-iit-a-[kwiita] _{pwd}			*		
b. kw-iit-a-[] _{pwd}		*W	L		
c. kw-iit-a-[iit-a] _{pwd}	*W		L		*W

What such patterns highlight is the need for the theory of reduplication to allow copying to access the entire string, not just the portion that the reduplicative affix scopes over. In our theory, this is encoded in the Separation assumption: morphological and phonological structure can diverge.

We want to reiterate the point made by Marantz (1987): these kinds of bracketing paradoxes argue against theories that architecturally intersperse phonological derivations with morphological affixation, such as Lexical Phonology and Morphology and its Stratal OT implementation (Kiparsky, 1982; Bermúdez-Otero, 2018, a.o.). For the Kihehe example, such a theory would apply reduplication to /tova/, yielding [tova-tova], and then attach the prefix /ku-/, yielding [ku-tova-tova]. We know that agreement prefixes are not reduplicated in these cases, so reduplication serves as a diagnostic for attachment height. How, then, can reduplication access /ku-/ in [kwiita-kwiita], if it has not been added yet?

Bracketing paradoxes are a general problem for a theory that targets morphosyntactic constituents for reduplicative doubling, as MDT does. To deal with the problem, Inkelas and Zoll (2005) introduce is a special constituent, the *Proot*, which is inspired by the morphological root but can incorporate material from other morphemes if phonologically required (Inkelas and Zoll 2005:140). The Proot comes with its own correspondence constraints, rankable separately from constraints on input-root correspondence—in a theory that otherwise denies the existence of identity requirements. It is hard to see the Proot as anything but a stipulation, since its only purpose and justification is to analyze bracketing paradoxes. It requires a view of the grammar where phonology can reassign morphological affiliations to segments based on syllabification or minimality requirements. By contrast, in our theory, there are no Proots, only actual roots and affixes plus derivational phonological copying.

5.3 Prosodic Circumscription

The next alternative we turn to is an older theory, known as Prosodic Circumscription (McCarthy and Prince 1990, Lombardi and McCarthy 1991, see McCarthy 2000 for a critique). This is a rule-based theory of Prosodic Morphology with operations that carve out a prosodic constituent X from an input Y and then manipulate either X (Positive Prosodic Circumscription) or the residue Y-X (Negative Prosodic Circumscription). Take, for example, infixing reduplication in Samoan, where the plural is a light syllable reduplicant prefixed to the main stress foot (a right-aligned moraic trochee):

- (67) Samoan infixing reduplication: prefix to stressed syllable (McCarthy and Prince 1990; Mosel and Hovdhaugen 1992; Zuraw et al. 2014, a.o.)

a.	tá:	‘strike’	<ta>tá:	‘strike pl.’
b.	alófa	‘love’	a<lo>lófa	‘love pl.’
c.	saváli	‘walk’	sa<va>váli	‘walk pl.’
d.	?anapógi	‘be fast’	?ana<po>pógi	‘be fast pl.’

The Prosodic Circumscription analysis of *alolofa* runs as follows (see (68)): the operation of circumscription identifies the main stress foot (*lofa*), removes the residue (*a*), prefixes a light syllable template ($\Sigma-$) to the foot (Φ) and fills it (*lo-lofa*), finally reinstating the residue *a-* (*alolofa*). In Lombardi & McCarthy’s elaboration of the theory, an edge parameter is added, specifying whether to take a leftmost or rightmost constituent. In Samoan it is the rightmost foot:

- (68) Positive prosodic circumscription in Samoan (McCarthy and Prince 1990:231)

$$\begin{aligned}
 \Sigma- : \Phi(\text{alofa}) &= \Sigma-(\text{alofa}:\Phi) * \text{alofa}/\Phi \\
 &= \Sigma-(\text{lofa}) * a \\
 &= \text{lolofa} * a \\
 &= \text{alolofa}
 \end{aligned}$$

McCarthy & Prince do discuss Yidj (1990:233–234), with the aim of accounting for it without also ruling in patterns that violate Moravcsik’s Generalization. McCarthy & Prince do not offer a complete formalized account, but they suggest Yidj circumscribes the first foot (=two syllables) of the base to form a minimal word, and reduplication then operates on the resulting constituent because it is specified as the domain of reduplication. This supposedly allows the Marantzian view of reduplication to stand: it is association to a template, rather than prosodic constituent copying. There is some resemblance between the circumscribed foot/word domain in Yidj and the Proot of MDT: both are intermediate between prosodic and morphological constituents, and both are motivated just to derive otherwise troublesome cases.

It is easy to show that the pieces of Prosodic Circumscription theory can be combined to generate syllable-copying reduplication, the existence of which we have been questioning throughout this paper. Thus, Lombardi and McCarthy (1991) use Negative Prosodic Circumscription to isolate the rightmost syllable in the word and operate on the remainder in Choctaw. It is only a small leap to imagine positively circumscribing the leftmost syllable, and reduplicating that in much the same way that the minimal word is supposedly reduplicated in Yidj:

- (69) Generating syllable copying in Prosodic Circumscription (hypothetical example)

	Pos.Circumscription (σ , L)	Affix σ -	Fill in template	Reinstate residue
pa.ti	pa	σ -pa	pa-pa	pa-pa.ti
pit.ku	pit	σ -pit	pit-pit	pit-pit.ku

It is not surprising that an operational theory overpredicts, since deriving a restricted typology is not an overt goal as it is in OT and HS. McCarthy (2000) himself offers a critical reevaluation of the various cases that have been analyzed in circumscriptional terms, and he rejects Negative Prosodic Circumscription altogether, while arguing that prosodic faithfulness offers a better handle on the Positive uses (using traditionally-described Yidj as his main argument). We go even further, rejecting prosodic faithfulness such as syllable copying.

5.4 A Learnability Explanation?

5.4.1 Overview

The final alternative we consider exploits learnability. The idea is that certain patterns are rare because they are difficult to learn. This removes the need for a restrictive theory, since the explanation for typological gaps does not reside in the theory alone. A variety of grammars are generated, and some are easier to learn than others. Learning can be difficult for several reasons: if the generalizations or interactions are particularly complex, or if the statistical evidence for crucial interactions is sparse (see Moreton 2008, Staubs 2014, Stanton 2016, Gouskova and Gallagher 2020, a.o.).

Reasoning about learnability relies on evidence from computational modeling and experimentation. Two experimental studies have examined syllable copying reduplication: Haugen et al. (2022) and Wang (2024). Both argue that syllable copying is learnable, but less so than other kinds of reduplication. Our take on the studies is that as designed, they fall short of supplying learnability evidence.

5.4.2 Haugen et al. (2022)

Haugen et al.'s (2022) experiment examined syllable-copying reduplication in the ease-of-learning artificial grammar paradigm: people saw the artificial language's plurals and augmentatives, and they were asked to choose the correct option. Participants were divided into three groups: CV-, CVC-, and CVX-. Each group saw a different partial reduplication pattern for the augmentative (see (70)). All three groups were simultaneously taught total reduplication for the plural, which everyone learned easily. Each response got corrective feedback. All speakers were English monolingual undergrads at the University of Arizona.

(70) Some sample items with expected responses from Haugen et al. (2022, supplementary materials)

	Base	Plural (all conditions)	Augmentative		
			CV cond.	CVC cond.	CVX cond.
a.	vamseta	vamseta-vamseta	va-vamseta	vam-vamseta	vam-vamseta
b.	nekola	nekola-nekola	ne-nekola	nek-nekola	ne-nekola
c.	nokrutu	nokrutu-nokrutu	no-nokrutu	nok-nokrutu	nok-nokrutu
d.	kutwipa	kutwipa-kutwipa	ku-kutwipa	kut-kutwipa	kut-kutwipa

Participants received feedback on all answers, so that the timecourse of learning could be analyzed. Haugen et al. found that the third pattern, CVX, is learned more slowly than a consistent CV or CVC pattern. The supposed interpretation is that syllable copying should be generated by the grammar; it is rare because it is harder to learn. We do not believe that this is the right interpretation of these results, however.

First, the evidence comes from English speakers, and English syllabification interacts with stress in complicated ways (Kahn 1976, Selkirk 1982, Hammond 1997, a.o.). The participants only saw orthographic stimuli, so nobody knows how they stressed these words. If they stressed the penult, as in [na.'kru.tu], then the usual rules of English syllabification would put [kɹ] in the onset. If they stressed the antepenult, as in ['nekələ], then the syllabification of [k] is debatable (ambisyllabic, etc.—see Kahn 1976 et seq.). Thus, both orthographic presentation and the use of English-speaking participants muddle the results; the CVX condition might have been difficult for reasons that have nothing to do with syllabification. In order to produce interpretable results, an artificial grammar experiment must control for the native language. Experiments that do control for native language sometimes report different outcomes depending on the participants' language background (cf. White et al. 2018, Conklin et al. 2024).

Second, there is an alternative interpretation of Haugen et al.'s result: participants developed an explicit strategy for what to copy, using metalinguistic generalizations about orthographic consonants and vowels (see sec. 5.4.4). The generalizations might go something like this:

- (71) CV condition: copy up to the first vowel letter.
- (72) CVC condition: copy up to the second consonant letter.
- (73) CVX condition: copy up to the first vowel if the word starts with CVCV, and copy up to the second consonant if the word starts with CVCCV.

The generalization for the CVX condition is undoubtedly more complicated, so it is no wonder it took participants longer to work it out. But it does not require a syllable-copying theory of reduplication to produce. Research on artificial grammar learning has repeatedly found differences between formally simple and formally complex generalizations (Moreton, 2012; Pater and Moreton, 2012, a.o.). Thus, in order to demonstrate that the syllable copying pattern is harder to learn in an artificial grammar, it needs to be presented in a way that does not facilitate orthography-based generalizations, and it needs to be compared to a pattern of equal logical complexity. For example, a CVX-prime condition could be added that inverts the generalization (CVC copying for CVCV-initial words, and CV copying for CVCCV-initial words). Confounds introduced by participants' native phonologies must also be ruled out.

5.4.3 Wang (2024)

Wang (2024) examined reduplication using artificial grammars in a poverty-of-the-stimulus paradigm (Wilson 2006; Maldonado and Culbertson 2022). In Wang's experiment, people heard ambiguous training data (e.g., [dɒv.gi] ~ [dɒv-'dɒv.gi], [ʃæp.mə] ~ [ʃæp-'ʃæp.mə]), which were consistent with multiple generalizations: "copy first CVC", "copy coronal-vowel-labial", "copy the first three segments", "copy first syllable", "copy first heavy syllable", etc. People then heard testing stimuli of various additional shapes ([av.di], [stæb.gə], [dɛ.ber], [tæ.preɪ])¹⁸, and produced pronunciations of "plurals". Like in Haugen's experiment, the participants were English speakers, leaving the experiment vulnerable to the same criticism (native language confounds). Unlike in Haugen et al.'s experiment, the stimuli were audio (pronounced by a synthesizer voice), so the stress locations were known.

Wang found "marginal support for the base-dependent syllable copying generalization" (2024 p. 49): the stimuli with a single C after the first syllable were more likely to be copied as vowel-final (e.g., [dɛ-'dɛber]) than [av.di] and [stæb.gə] ([av-'av.di], [stæb-'stæb.gə]). Not all ...VCCV... stimuli patterned together, however: [fædnɒʊ], [tæpreɪ], [ʃipnɛɪ] were not significantly different from [dɛ.ber]. By contrast, there was a much clearer effect for word-initial onsets: these were usually copied faithfully ([stæb-'stæb.gə] not [tæb-'stæb.gə], etc.). Wang interprets both the rime disparities and the onset copying as base-dependent: the syllable structure of the base determines what is copied.

Wang's experiment is somewhat exploratory in design, since participants were free to produce any forms (some did not even generate reduplication). We are once again skeptical, however, that the syllable copying explanation is the right one. Given the heterogeneous patterning of the ...VCCV... stimuli, other explanations suggest themselves. Wang (2024) does not list all the stimuli, but the ones that are discussed show varying degrees of similarity to existing English words (*stab*, *fad*, *sheep*, *of*). This could affect the likelihood of copying beyond any phonological/typological explanations. Wang also finds an effect (attributed to substantive grounding) of the sequences at the boundary between the reduplicant and the base, which affect whether a C is copied. Since the clusters created by either copying or omitting the coda consonant were not tightly controlled, this could well be an item-level effect rather than something deeper about the grammatical nature of reduplication. Finally, and most importantly, the nature of this experiment opens up the possibility that speakers treated it more like a language game than a grammatical task, as we discuss below.

¹⁸The transcriptions and syllabification are Wang's, and once again the syllabification is debatable. The [p] in [tæpreɪ] is in a position that some would analyze as ambisyllabic (cf. Kahn (1976, p. 84) on the ambisyllabicity of [t] in *mattress*).

5.4.4 Must the Theory Generate Syllable Copying?

There is evidence that people can learn typologically unattested patterns in an artificial grammar experiment. Much work that specifically tests typologically attested and unattested patterns finds a difference of degree rather than an utter failure to learn the unattested pattern. For example, Moreton et al. (2020) taught participants to recognize words generated either via total reduplication (*kolive-su-kolive*) or via syllable reversal (*kolive-su-veliko*). Total morpho-phonological reduplication is typologically ubiquitous, while reversal is unattested except in language games (see Moreton et al. 2020). In the experiments, the total reduplication group only attained at most 80% accuracy,¹⁹ which varied based on whether they worked out an explicit rule. The reversal group attained a maximum 60% accuracy (based on their Fig. 2, p. 5). Strikingly, *people with an explicit rule for typologically unattested reversal were more accurate by the end of the experiment than people who lacked an explicit rule for typologically ubiquitous reduplication*.

Many artificial grammar experiments do not debrief their participants on awareness of an explicit strategy, but this might reveal whether people are using explicit rules or some implicit acquisition mechanism similar to what children use. We know that language games and secret languages manipulate segments, syllables, and feet (Bagemihl 1989, Demisse and Bender 1983, Cowan et al. 1985, Ito et al. 1996, Barlow 2001, a.o.). Language games, however, rely to a large extent on metalinguistic awareness. For example, Cowan (1989) shows that Pig Latin is learned only with the onset of literacy, probably because children teach each other rules in orthographic terms (“move the first consonant to the end and add *-ay*”, thus *bug* → *ug-bay*). It may be the case that in both Wang’s and Haugen et al.’s studies of reduplication, speakers sought similar generalizations.

Zooming out from the experiments, though, we could ask how speakers can possibly construe an explicit rule for, say, syllable reversal (as in a Saramaccan language game, see Bagemihl 1989) if the grammar does not generate syllable copying or movement. The conclusion has to be that language games and explicit strategies employed by artificial grammar participants can manipulate linguistic objects of which speakers have conscious awareness. Speakers of various languages can have metalinguistic awareness of syllables, segments, and even morae (often taught explicitly during reading instruction). This knowledge can be accessed to create rules that are not grammatical in nature. To clarify, we are certainly not questioning the existence of syllables. But their role in phonology appears to be more asymmetrical than suggested in theories such as Prosodic Circumscription and the original STS of McCarthy et al. (2012). In particular, no phonological operations/faithfulness constraints refer to syllables, though markedness constraints do.

To summarize, we are sympathetic to learnability-based explanations for differences between rare and amply attested patterns: perhaps rare patterns are harder to learn. But we also believe that a theory should not generate unattested patterns. We have been questioning the existence of syllable copying reduplication, which is thought to be rare, and arguing instead that it is non-existent. Thus, a grammar should not be able to generate it. Certainly it should not be as easy as applying a COPY-SYLL operation or ranking STROLE high.

6 Loose Ends

This section addresses some issues of more general interest, beyond Yidjn. We discuss a couple more cases that have been characterized in syllable-copying terms, and we wrap up with a brief discussion of infixation pivots.

¹⁹Compare this to near-ceiling accuracy across all conditions for total reduplication in Haugen et al. (2022). The reason is likely the nature of the task: learning to recognize a static pattern (as in Moreton et al.) is harder than learning a morphological affixation rule (as in Haugen et al.). There are also differences between participants tested on the internet vs. in-person in the lab, and between platforms such as Mechanical Turk vs. Prolific.

6.1 Other cases of syllable copying?

We now turn to other cases that have been analyzed in terms of syllable copying. Haugen and Hicks Kennard (2011) and McCarthy et al. (2012) list Hiaki (a.k.a. Yaqui, Uto-Aztecan) as an example of syllable copying reduplication: the basic pattern is described as copying CVC- only when the base has a closed syllable. We do not think this stands up to scrutiny. Harley and Florez Leyva (2009) conclude that the extent of reduplicative copying in Hiaki has to be at least partially lexically specified (see also Haugen 2014). Reduplication, which carries a number of meanings, varies in form between CV-, CVCV-, and CVC- (see (74)). According to Harley & Florez Leyva's quantitative survey, 67% of verbs reduplicate CV-, and the remainder follow other patterns. Moreover, multiple patterns can apply to the same root: *nooka* 'talk' reduplicates as *no-noka* 'pl. subj' and *non-noka* 'emphasis'. Thus, we see counterexamples in both directions. There may be some tendencies toward which shapes of roots follow which pattern, but the patterns can profitably be analyzed as suppletive allomorphy (or just unrelated morphemes expressed through reduplication, with different templates). There is no need to invoke syllable copying here.

(74) A few subpatterns in Hiaki habitual reduplication (Harley and Florez Leyva 2009)

a.	CV-	bwiika	bwi-bwika	'sing'
b.		hinu	hi-hinu	'buy'
c.	CVCV-	kitok-te	kito-kitokte	'contract muscle.intr'
d.	CVC-	chamta	cham-chamta	'mash'
e.		kiimu	kik-kimu	'enter'
f.		chepte	che-chepte	'jump pl. subj'
g.	doublets:	chepte	chep-chepte	'jump (habitual)'

McCarthy et al.'s (2012) main examples of syllable vs. segment-level copying are Manam and Balangao: both have foot-sized reduplicants, but Manam supposedly copies at the syllabic level, while Balangao copies segments.

(75) Manam vs. Balangao, per McCarthy et al. (2012)

a. Manam: syllable copying			b. Balangao: segment copying		
salága	salaga-lága	'long'	ʔuma	ka-ʔuma-ʔuma	'always making fields'
moíta	moita-íta	'cone shell'	ʔabulot	ka-ʔabu-ʔabulot	'believers of everything'
malabónj	malabom-bónj	'flying fox sp.'	taynan	ma-tayna-taynan	'repeatedly be left behind'

These languages both copy a foot-sized unit, but they differ in several respects that deflate the argument for a syllable- vs. segment-level copying analysis. For one thing, Manam is suffixing, while Balangao is prefixing. It is clear that Balangao is not copying syllables: if it were, *taynan* would be copied whole.²⁰ But is it far from clear that Manam is copying syllables rather than segments (and indeed McCarthy et al. do not have a ranking argument for *COPY-SEG ≫ *COPY-σ).²¹ Manam's phonotactics and stress system do not offer any good options besides *salaga-laga* and *malabom-bonj*. Manam allows only nasals in word-final codas (Lichtenberk 1983), so options such as *salaga-lag* are ruled out phonotactically (CODACOND,

²⁰Balangao has more than one type of reduplication: the disyllabic CV(C)CV type is usually mentioned in the literature, but there is also CVC and CV reduplication (see Shetler 1976). Stress is contrastive on the last two syllables, but not shown in the examples in the source.

²¹Another case where differential syllable- vs. segment-level copying has been used in STS is Mbe (Lamont 2023). This case is too complex to reanalyze in detail here, but there does not seem to be straightforward arguments for syllable-level copying in that analysis, either.

Ito 1986). In C-final stems such as *malabón*, copying just one heavy syllable is the best way to satisfy the template and keep to the language's stress preferences (ultima if heavy, else penult). The WEIGHT-TO-STRESS PRINCIPLE (WSP, "No unstressed heavy syllables", Prince 1990) favors the shorter H reduplicant in this trochaic system. Thus, we can easily analyze Manam assuming only a COPY-SEG operation.

(76) Manam analyzed without syllable copying: *salaga-laga*

	/salága-[] _{Ft} /	HD(Ft)	*COPY-SEG	CODA COND	IDENT[place]
a. →	salága-[lága] _{Ft}		*		
b.	salága-[lág] _{Ft}		*	*W	
c.	salága-[] _{Ft}	*W	L		

(77) Manam monosyllabic reduplicants favored by the WEIGHT-TO-STRESS PRINCIPLE

	/malabón-[] _{Ft} /	HD(Ft)	*COPY-SEG	CODA COND	IDENT[place]	WSP
d. →	malabón-[bón] _{Ft}		*			
e.	malabón-[lábon] _{Ft}		*			*W

(Next step: [malabom-bon] by CODA COND ≫ IDENT[place], not shown)

The one case that we identified that looks the closest to syllable copying is Madurese, and this turns out to not be syllable copying when examined in more detail. This case is interesting not only because it appears to copy a syllable in its entirety but also because the final syllable is copied into a prefixing reduplicant:

(78) Madurese right-edge reduplication (Stevens 1968, Davies 2010): a violation of Marantz's and Moravcsik's Generalizations?

a.	bu.ku	'book'	ku-bu.ku	'books'
b.	po.te	'white'	te-po.te	'very white'
c.	to.les	'write'	les-to.les	'write (pl)'
d.	pen.ter	'smart'	ter-pen.ter	'very smart'
e.	ma.en	'play'	en-ma.en	'play (pl)'
f.	pra.o	'ship'	o-pra.o	'ships'
g.	tap.pla	'tablecloth'	pla-tap.pla	'tablecloths'
h.	sem.prot	'spray'	prot-sem.prot	'spray (pl)'
i.	pot.tre	'princess'	tre-pot.tre	'princesses'

Madurese appears to violate both Moravcsik's and Marantz's Generalizations, the latter of which encodes the tendency for prefixes to copy from the left and suffixes from the right. In her extensive study of purported counterexamples to Marantz's Generalization, Nelson (2003:16-17) argues that Madurese (and related Malay) reduplication results from deletion of contiguous material to reduce the reduplicant to one syllable—a rule that is also observed in compounds. As for Moravcsik's Generalization, a closer look at the pattern shows that copying does not target a syllable. As shown in (79a-c), when the root is followed by a vowel-initial suffix, its final consonant is syllabified into an onset, and yet that consonant is copied into the reduplicant. The suffix is indeed present when reduplication applies, since in [bu.waʔ.an], the glottal stop—an obligate coda—is epenthesized to avoid hiatus of identical vowels, **a-a*, and [ʔ] is copied into the reduplicant.

(79) Right-edge syllabification mismatches (Davies 2010:131)

a.	/tamen-an/	ta.me.nan	'plant (N)'	men-ta.me.nan	'plants'
b.	/pekker-an/	pek.ke.ran	'think and think'	ker-pek.ke.ran	'think and think'
c.	/kerem-e/	ke.re.me	'send to Loc.'	rem-ke.re.me	'send to Loc (pl)'
d.	/buwa-an/	bu.waʔ.an	'fruit'	waʔ-bu.waʔ.an	'fruits'; cf. [buwa] 'fruit'

While it is not a counterexample to Moravcsik’s Generalization, Madurese is important in that it lends support to our assumptions of Separation and Serialism. Relevant to Separation, it is yet another bracketing paradox. As Davies notes, [men_{PI}-[tamen_V-an_N]] ‘plants’ is a plural of the noun, so it should be expected to copy the nominalizing suffix—thus the reduplicative affix attaches to the branching deverbal nominal [tamen-an], and yet the reduplicant does not copy the suffix. Relevant to Serialism, Madurese demonstrates the need for copying to be derivationally interspersed with other phonological operations. Glottal stop epenthesis enforces a surface-true phonotactic constraint, and epenthesis must apply before copying.

6.2 What about syllables as infixation pivots?

We have questioned the existence of syllable copying in reduplication, taking an essentially conservative position that Moravcsik’s Generalization is a universal that the theory must derive. But this raises the question of what role syllables play elsewhere in Prosodic Morphology²²—in particular, what about infixation? According to the dominant theory of infixation, infixes have phonological positioning requirements known as “pivots”. The pivots have been variously argued to be phonological/prosodic constituents and arbitrary C/V frames (see Yu 2003, McCarthy 2003b, Kalin 2022). Is the syllable a possible pivot?

There are significant asymmetries in the typological attestation of some pivots vs. others (see Mooney 2023a). It is common for infixes to precede or follow stressed syllables: the reduplicative infix in Samoan prefixes to the word-final trochaic foot (*a<lo>(lófa* ‘love pl.’, (67)), and in Ulwa, the segmental infix *-ka-* suffixes to the word-initial iambic foot (see (80)). It is also common for infixes to appear close to the left edge but not quite initially, e.g., preceded by a consonant or appearing right before the first vowel: Tagalog *-um-* freely varies between these options (see (81)). Crucially, Ulwa would be difficult to describe if syllabification were not assumed; the infix appears not just after the stressed vowel but after the stressed syllable (e.g., why *ka.rás.<ka>mak* and not *ka.rá.<ka>s.mak*).

(80) Ulwa *-ka-*: a suffix to the left-aligned iambic foot (McCarthy and Prince 1990, Green 1999)

	Bare N	Someone’s N	
a.	(bás)	bás<ka>	‘hair’
b.	(ki:)	kí:<ka>	‘stone’
c.	(sapá:)	sapá:<ka>	‘forehead’
d.	(sú:) lu	sú:<ka>lu	‘dog’
e.	(bás)karna	bás<ka>karna	‘comb’
f.	(aná):la:ka	aná:<ka>la:ka	‘chin’
g.	(karás)mak	karás<ka>mak	‘knee’ not *kará<ka>smak

(81) Tagalog *-um-* (McCarthy 2003b & refs therein)

a.	sulat	s<um>ulat	‘to write’
b.	?abot	?<um>abot	‘to reach for’
c.	preno	pr<um>eno ~ p<um>reno	‘to brake’
d.	gradwet	gr<um>adwet ~ g<um>radwet	‘to graduate’

²²Prosodic Morphology also concerns itself with truncation and hypocoristics (e.g., *psychology*→*psych*, *Joseph*→*Joe*). The language’s constraints on syllables clearly play a role, but as a target rather than in a base-dependent way (see Weeda 1992; Kenstowicz 1994; Wiese 2001, a.o.), much as we have been arguing for Yidj. Furthermore, it is controversial whether hypocoristic formation falls into the same class of phenomena as reduplication and infixation, since it does not appear to spell out morphosyntactic features (Bye and Svenonius 2012:4). It creates allomorphs of roots, and it might reference prosodic constituents much as secret languages do, but whether it is a core part of the morphophonological grammar is an open question.

Positioning requirements are usually formalized as Alignment constraints or subcategorization frames (see McCarthy and Prince 1993a, Kalin and Rolle 2023 for examples). Either theory of pivots combines phonological structures (C, V, syllable, foot, etc.) and edges (L, R) freely, which would predict that an infix can be positioned after the initial syllable of the word, or before the final syllable. And yet Yu (2007) finds no example of the first syllable as the pivot. This is surprising since infixes that are left-tropic are more common than right-tropic ones, if the latter exist at all (Mooney 2023a shows that right-tropic infixes are actually prominence-tropic). The only purported counterexample that has emerged since Yu's work is Yeri (Torricelli, Papua New Guinea, Wilson 2014, Wilson 2017). As examples (a–e) show, the <me/mi/m> imperfective infix appears after the first syllable, respecting the presence of a coda but not the position of stress (marked as in sources). There is also a <pe/pi/p> additive infix with the same distribution.

(82) Yeri infixes (Wilson 2014, 2017)

	<i>stem</i>	<i>imperfective</i>	
a.	or	or.me	'sleep'
b.	'al.mo	al.me.mo	'die'
c.	'ol.bil	ol.me.bil	'enter'
d.	a.'sol.kia	a.mi.'sol.kia, am.'sol.kia	'do badly'
e.	a.'ni.bir	a.mi.'ni.bir, am.'ni.bir	'send'
f.	do.di	do.mé.di, dómdi	'stand'

Just as Yidj, this rare counterexample to an otherwise strong generalization invites a critical look, and the details cast doubt on the syllable-based generalization. First, Wilson (2017) cites no examples of infixing into words with other syllable types in the relevant position, even though the language admits [pr, pl, kr, kl, dr, sp, sk] as onset clusters (Wilson 2017 p. 68). The only examples where the [me/mi] allomorphs suffix to a coda have [l, r] in that position. Second, in characterizing the distribution of the [m] allomorph, Wilson suggests that velars and labials delete after the infix, though this rule has exceptions. We think there is a different generalization that is actually more consistent with the facts: the [m] form results from overwriting the first consonant of the stem that is neither initial nor coronal (see (83)). This is a distribution more characteristic of a floating feature, and there is no shortage of analyses (Zoll 1997, 1998; Wolf 2007, a.o.). In sum, this case raises some questions for future investigation, but we do not think it provides a clear-cut example of infixation after the first syllable.²³

(83) Yeri complications (Wilson 2017)

	<i>stem</i>	<i>imperfective</i>	
a.	dal.ko.ki	dal.mo.ki	'look for'
b.	dar.ko.ka	dar.me.ko.ka	'rush'
c.	a.wil	a.mil	'hang'
d.	o.gi.wa	o.mi.wa	'ask'
e.	no.bia	no.me.bia	'talk'
f.	gal.pi.sia	gal.bi.sia	'separate' fr. /gal-m-pisia/
g.	o.pe.ri	ombe.ri	'pour' om.be.ri? o.mbe.ri?

Even setting aside the question of the initial syllable as a pivot, there are undeniable gaps in the typology of infixation (see Mooney 2023a). For example, there are few examples of suffixing to the first vowel, and they are open to alternative interpretations (Yu 2007:101). The surveys of infixation also frequently conflate reduplicative and segmental infixes, despite noted asymmetries in their patterning (see McCarthy 2000). If these typological generalizations hold up, they cast doubt on subcategorization frames/Alignment

²³For more discussion of the typology of infixation pivots, see Samuels (2010) and references therein.

as theories of infixation. A prominent alternative in the literature is that infixation is incidental to some other requirement. For example, infixation can be driven by a requirement that the word start with a segment from the root (Hendricks 1999, Crowhurst 2004), or that the word's overall shape obey rhythmic constraints (see, e.g., McCarthy 1982 on expletive infixation in English). Our own proposal suggests yet another possibility: the distribution of infixes is determined in part by the properties of the operation that positions them, and that operation might not be able to refer to syllables, much as copying operations.

7 Conclusion

In this paper, we revisited Moravcsik's (1978) Generalization: reduplication does not copy constituent structure such as syllables from the base. The best purported example of reduplicative syllable copying, Yidj, turns out to have more counterexamples to the description in the reduplication literature than examples consistent with the description. We demonstrated that a simplified version of Serial Template Satisfaction theory with only morpheme- and segment-based copying is sufficient to analyze the full pattern. Moreover, there are features of Yidj reduplication that, we think, point toward favoring STS over alternatives such as Base-Reduplicant Correspondence Theory, which continues to remain popular. BRCT cannot easily handle several of the opaque patterns of reduplication, such as those in apocopating roots. We also argued that Morphological Doubling Theory, with its assumptions about sign-based morphology and cophonologies, has difficulties with multiple aspects of the pattern: defaulting to the σ CV pattern for longer roots, and bracketing paradoxes in verbal reduplication (both in Yidj and in languages such as Kihehe).

We argue that the right theory of reduplication must be serial to accommodate opacity. It also must be template-based, rather than allowing markedness full reign in determining reduplicant shape (as in MDT and the Generalized Template Theory version of BRCT). Finally, in order to handle bracketing paradoxes, reduplication must happen in the phonological grammar after the full morphological structure has been linearized; in this, we echo earlier arguments that morphological and phonological structure are not coextensive.

References

- Arad, Maya. 2003. Locality constraints on the interpretation of roots: The case of Hebrew denominal verbs. *Natural Language and Linguistic Theory* 21:737–778.
- Austin, Peter. 1981. *A Grammar of Diyari, South Australia*. Cambridge: Cambridge University Press.
- Bagemihl, Bruce. 1989. The Crossing Constraint and 'backwards languages'. *Natural Language & Linguistic Theory* 7:481–549.
- Barlow, Jessica A. 2001. Individual differences in the production of initial consonant sequences in pig latin. *Lingua* 111:667–696.
- Benua, Laura. 1997. Transderivational identity: Phonological relations between words. Doctoral Dissertation, University of Massachusetts, Amherst, MA.
- Bermúdez-Otero, Ricardo. 2018. Stratal phonology. In *The Routledge handbook of phonological theory*, ed. S.J. Hannahs and Anna R.K. Bosch, 100–134. Abingdon: Routledge.
- Bye, Patrik, and Peter Svenonius. 2012. Non-concatenative morphology as epiphenomenon. In *The Morphology and Phonology of Exponence*, ed. Jochen Trommer, 427–495. Oxford University Press.
- Caballero, Gabriela. 2006. "Templatic backcopying" in Guarijio abbreviated reduplication. *Morphology* 16:273–289.
- Chomsky, Noam, and Morris Halle. 1968. *The Sound Pattern of English*. New York: Harper & Row.

- Coetzee, Andries, and Joe Pater. 2011. The place of variation in phonological theory. In *The Handbook of Phonological Theory*, ed. John Goldsmith, Jason Riggle, and Alan Yu, 401–434. Oxford: Blackwell, 2 edition.
- Conklin, Jenna, Elliot Stork, Ruken Bastimar, and Xueqi He. 2024. Impacts of direction and morphological locus of control on learnability of sibilant harmony. *Laboratory Phonology* 15.
- Cowan, N., M. D. S. Braine, and L. A. Leavitt. 1985. The phonological and metaphonological representation of speech: Evidence from fluent backward talkers. *Journal of Memory and Language* 24:679–698.
- Cowan, Nelson. 1989. Acquisition of pig latin: a case study. *Journal of child language* 16:365–386.
- Crowhurst, Megan. 2004. Mora Alignment. *Natural Language & Linguistic Theory* 22:127–177.
- Davies, William D. 2010. *A grammar of Madurese*, volume 50. Walter de Gruyter.
- Demisse, T., and M. L. Bender. 1983. An argot of Addis Ababa unattached girls. *Language in Society* 12:339–347.
- Dixon, R.M.W. 1972. *The Dyirbal language of North Queensland*. Cambridge: Cambridge University Press.
- Dixon, R.M.W. 1977a. *A Grammar of Yidiny*. Cambridge: Cambridge University Press.
- Dixon, R.M.W. 1977b. Some phonological rules of Yidiny. *Linguistic Inquiry* 8:1–34.
- Elfner, Emily. 2016. Stress-epenthesis interactions in Harmonic Serialism. In *Harmonic Grammar and Harmonic Serialism*, ed. John J. McCarthy and Joe Pater, 261–300. London: Equinox Publishing.
- Fenger, Paula, Maria Kouneli, and Jonathan David Bobaljik. 2025. Dominant domains in vowel harmony: A structural approach to a linear asymmetry. *Natural Language & Linguistic Theory* 43:2771–2820.
- Gafos, Adamantios. 1998. A-templatic reduplication. *Linguistic Inquiry* 29:515–527.
- Ghomeshi, Jila, Ray Jackendoff, Nicole Rosen, and Kevin Russell. 2004. Contrastive focus reduplication in English (the salad-salad paper). *Natural Language & Linguistic Theory* 22:307–357.
- Gouskova, Maria. 2007. The Reduplicative template in Tonkawa. *Phonology* 24:367–396.
- Gouskova, Maria, and Gillian Gallagher. 2020. Inducing nonlocal constraints from baseline phonotactics. *Natural Language & Linguistic Theory* 77–116. URL <https://doi.org/10.1007/s11049-019-09446-x>.
- Gouskova, Maria, and Juliet Stanton. 2021. Learning complex segments. *Language* 97:151–193. URL <https://ling.auf.net/lingbuzz/004815>.
- Green, Thomas Michael. 1999. A lexicographic study of Ulwa. Doctoral dissertation, Massachusetts Institute of Technology.
- Hale, Kenneth Locke. 1959. A Papago grammar. Doctoral dissertation, Indiana University.
- Halle, Morris, and Alec Marantz. 1993. Distributed Morphology. In *The View from Building 20: Essays in Honor of Sylvain Bromberger*, ed. Kenneth Hale and Samuel Jay Keyser, 111–176. Cambridge, MA: MIT Press.
- Hammond, Michael. 1997. Vowel quantity and syllabification in English. *Language* 73:1–17.
- Harley, Heidi, and Maria Florez Leyva. 2009. Form and meaning in Hiaki (Yaqui) verbal reduplication. *International Journal of American Linguistics* 75:233–272.
- Haude, Katharina. 2006. A grammar of Movima. Doctoral dissertation.
- Haugen, Jason D. 2014. Hiaki (Yaqui) reduplication redux: Morphological and prosodic targets in syllabic reduplication. *International Journal of American Linguistics* 80:507–532.
- Haugen, Jason D, and Cathy Hicks Kennard. 2011. Base-dependence in reduplication. *Morphology* 21:1–29.
- Haugen, Jason D., Adam Ussishkin, and Colin Reimer Dawson. 2022. Learning a typologically unusual reduplication pattern: An artificial language learning study of base-dependent reduplication. *Morphology* 32:299–315.

- Hayes, Bruce. 1995. *Metrical Stress Theory: Principles and Case Studies*. Chicago: The University of Chicago Press.
- Hayes, Bruce. 1999. Phonological restructuring in Yidiny and its theoretical consequences. *The derivational residue in phonological Optimality Theory* 175–295.
- Hayes, Bruce, and May Abad. 1989. Reduplication and syllabification in Ilokano. *Lingua* 77:331–374.
- Hendricks, Sean. 1999. Reduplication without Templates: A Study of Bare-Consonant Reduplication. Doctoral Dissertation, University of Arizona, Tucson, AZ.
- Hicks-Kennard, Catherine. 2004. Copy but don't repeat: the conflict of dissimilation and reduplication in the Tawala durative. *Phonology* 21.
- Inkelas, Sharon, and Cheryl Zoll. 2005. *Reduplication: doubling in morphology*. Cambridge: Cambridge University Press.
- Ito, Junko. 1986. Syllable Theory in Prosodic Phonology. Doctoral Dissertation, University of Massachusetts, Amherst. Published 1988. Outstanding Dissertations in Linguistics series. New York: Garland.
- Ito, Junko, Yoshihisa Kitagawa, and Armin Mester. 1996. Prosodic faithfulness and correspondence: Evidence from a Japanese Argot. *Journal of East Asian Linguistics* 5:217–294.
- Jaker, Alessandro, and Paul Kiparsky. 2020. Level ordering and opacity in Tetsót'iné: A Stratal OT account. *Phonology* 37:617–655.
- Kahn, Daniel. 1976. Syllable-based Generalizations in English Phonology. Doctoral Dissertation, MIT, Cambridge, MA. Published by Garland Press, New York, 1980.
- Kalin, Laura. 2022. Infixes really are (underlyingly) prefixes/suffixes: Evidence from allomorphy on the fine timing of infixation. *Language* 98:641–682.
- Kalin, Laura, and Nicholas Rolle. 2023. Deconstructing subcategorization: Conditions on insertion versus conditions on position. *Linguistic Inquiry*.
- Kastner, Itamar. 2019. Templatic morphology as an emergent property. *Natural Language & Linguistic Theory* 37:571–619.
- Kenstowicz, Michael. 1994. *Phonology in Generative Grammar*. Oxford: Blackwell.
- Kenstowicz, Michael. 1996. Base-Identity and Uniform Exponence: Alternatives to Cyclicity. In *Current Trends in Phonology: Models and Methods*, ed. Jacques Durand and Bernard Laks, 363–393. Manchester, England: European Studies Research Institute, University of Salford.
- Kimper, Wendell. 2011. Locality and globality in phonological variation. *Natural Language & Linguistic Theory* 29:423–465.
- Kiparsky, Paul. 1982. Lexical phonology and morphology. In *Linguistics in the Morning Calm*, ed. In-Seok Yang, volume 2, 3–91. Seoul: Hanshin.
- Kirchner, Jesse Saba. 2010. Minimal reduplication. Doctoral Dissertation, UC Santa Cruz, Santa Cruz, CA.
- Klein, Harriet B. 2005. Reduplication revisited: Functions, constraints, repairs, and clinical implications. *American Journal of Speech-Language Pathology* 14.
- Lamont, Andrew. 2023. Serial reduplication is empirically adequate and typologically restrictive. *Linguistic Inquiry* 54:797–839.
- Lichtenberk, Frantisek. 1983. *A Grammar of Manam*. Honolulu: University of Hawaii Press.
- Lombardi, Linda, and John J. McCarthy. 1991. Prosodic circumscription in Choctaw morphology. *Phonology* 8:37–71.
- Maldonado, Mora, and Jennifer Culbertson. 2022. Person of interest: Experimental investigations into the learnability of person systems. *Linguistic Inquiry* 53:295–336.
- Marantz, Alec. 1982. Re Reduplication. *Linguistic Inquiry* 13:483–545.
- Marantz, Alec. 1987. Phonologically induced bracketing paradoxes in full morpheme reduplication. In *The Proceedings of the West Coast Conference on Formal Linguistics 6*, ed. Megan Crowhurst, 203–212. Stanford: Stanford Linguistic Association.

- Marantz, Alec. 2013. Locality domains for contextual allomorphy across the interfaces. In *Distributed morphology today: Morphemes for Morris Halle*, ed. Ora Matushansky and Alec Marantz, 95–115. MIT Press.
- Marvin, Tatjana. 2002. Topics in the stress and syntax of words. Doctoral Dissertation, MIT, Cambridge, MA.
- McCarthy, John. 1982. Prosodic structure and expletive infixation. *Language* 58:574–590.
- McCarthy, John J. 1979. Formal problems in Semitic phonology and morphology. Doctoral Dissertation, MIT. Published by Garland Press, New York, 1985.
- McCarthy, John J. 2000. Faithfulness and prosodic circumscription. In *Optimality Theory: Phonology, Syntax, and Acquisition*, ed. Joost Dekkers, Frank van der Leeuw, and Jeroen van de Weijer, 151–189. Oxford: Oxford University Press.
- McCarthy, John J. 2002. *A thematic guide to Optimality Theory*. Cambridge: Cambridge University Press.
- McCarthy, John J. 2003a. Comparative markedness. *Theoretical Linguistics* 29:1–51.
- McCarthy, John J. 2003b. OT constraints are categorical. *Phonology* 20:75–138.
- McCarthy, John J. 2008a. The gradual path to cluster simplification. *Phonology* 25:271–319.
- McCarthy, John J. 2008b. The serial interaction of stress and syncope. *Natural Language and Linguistic Theory* 26:499–546.
- McCarthy, John J., Wendell Kimper, and Kevin Mullin. 2012. Reduplication in harmonic serialism. *Morphology* 22:173–232.
- McCarthy, John J., and Alan Prince. 1990. Foot and word in prosodic morphology: The Arabic broken plural. *Natural Language & Linguistic Theory* 8:209–283.
- McCarthy, John J., and Alan Prince. 1993a. Generalized Alignment. In *Yearbook of Morphology*, ed. Geert Booij and Jaap van Marle, 79–153. Dordrecht: Kluwer.
- McCarthy, John J., and Alan Prince. 1993b. *Prosodic morphology I: Constraint interaction and satisfaction*. New Brunswick: Rutgers University Center for Cognitive Science. Available as ROA-482 on the Rutgers Optimality Archive, <http://roa.rutgers.edu>.
- McCarthy, John J., and Alan Prince. 1994a. The emergence of the unmarked: Optimality in prosodic morphology. In *Proceedings of the North East Linguistic Society 24*, ed. Merce Gonzalez, 333–379. Amherst, MA: GLSA Publications.
- McCarthy, John J., and Alan Prince. 1994b. Two lectures on Prosodic Morphology (Utrecht, 1994). Part I: Template form in Prosodic Morphology. Part II: Faithfulness and reduplicative identity. Available as ROA-59 on the Rutgers Optimality Archive, <http://roa.rutgers.edu>.
- McCarthy, John J., and Alan Prince. 1995. Faithfulness and Reduplicative Identity. In *Papers in Optimality Theory*, ed. Jill N. Beckman, Laura Walsh, and Suzanne Urbanczyk, University of Massachusetts Occasional Papers 18, 249–384. University of Massachusetts, Amherst: GLSA Publications.
- McCarthy, John J., and Kathryn Pruitt. 2013. Sources of phonological structure. In *Linguistic Derivations and Filtering: Minimalism and Optimality Theory*, ed. Hans Broekhuis and Ralf Vogel, 110–135. London: Equinox Publishing.
- Mooney, Kate. 2023a. Asymmetric infixation. In *Proceedings of the Annual Meetings on Phonology*.
- Mooney, Kate. 2023b. The theory and typology of phonological reordering. Doctoral dissertation, New York University, New York, NY.
- Moravcsik, Edith. 1978. Reduplicative constructions. In *Universals of Human Language*, ed. Joseph Greenberg, 297–334 (vol. 3). Stanford: Stanford University Press.
- Moreton, Elliott. 1996/1999. Non-computable functions in Optimality Theory. Written in 1996. Revised and placed on Rutgers Optimality Archive in 1999.
- Moreton, Elliott. 2008. Analytic bias and phonological typology. *Phonology* 25:83–127.
- Moreton, Elliott. 2012. Inter-and intra-dimensional dependencies in implicit phonotactic learning. *Journal of memory and language* 67:165–183.

- Moreton, Elliott, Brandon Prickett, Katya Pertsova, Josh Fennell, Joe Pater, and Lisa Sanders. 2020. Learning repetition, but not syllable reversal. In *Proceedings of the Annual Meetings on Phonology*.
- Mosel, Ulrike, and Even Hovdhaugen. 1992. *Samoan reference grammar*. London, Tromsø: Oxford University Press.
- Nash, David. 1979. Yidiny stress: A metrical account. *CUNY Forum* 7/8:112–30.
- Nash, David. 1980. Topics in Warlpiri Grammar. Doctoral Dissertation, MIT, Cambridge, MA. Published by Garland Press, New York, 1986.
- Nelson, Nicole. 2003. Asymmetric Anchoring. Doctoral Dissertation, Rutgers University.
- Nyamahanga, Richard Mathias. 2025. Aspects of the phonology of Kihehe. Doctoral dissertation, Indiana University, Bloomington.
- Odden, David, and Mary Odden. 1985. Ordered reduplication in Kihehe. *Linguistic Inquiry* 16:497–503.
- Oltra Massuet, Maria Isabel. 1999. On the notion of theme vowel: a new approach to Catalan verbal morphology. Master's thesis, Massachusetts Institute of Technology.
- Pater, Joe. 2006. The locus of exceptionality: Morpheme-specific phonology as constraint indexation. In *Papers in Optimality Theory III*, ed. Leah Bateman, Michael O'Keefe, Ehren Reilly, and Adam Werle, 259–296. Amherst, MA: GLSA.
- Pater, Joe. 2008. Morpheme-specific phonology: Constraint indexation and inconsistency resolution. In *Phonological argumentation: Essays on Evidence and motivation*, ed. Steve Parker, 123–154. London: Equinox.
- Pater, Joe, and Elliott Moreton. 2012. Structurally biased phonology: Complexity in learning and typology. *Journal of the English and Foreign Languages University, Hyderabad* 3:1–44.
- Prince, Alan. 1990. Quantitative consequences of rhythmic organization. In *Parasession on the Syllable in Phonetics and Phonology*, ed. M. Ziolkowski, M. Noske, and K. Deaton, 355–398. Chicago: Chicago Linguistic Society.
- Pruitt, Kathryn. 2010. Serialism and locality in constraint-based metrical parsing. *Phonology* 27:481–526.
- Riggle, Jason. 2006. Infixing reduplication in Pima and its theoretical consequences. *Natural Language and Linguistic Theory* 24:857–891.
- Samuels, Bridget. 2010. The topology of infixation and reduplication. *Linguistic review* 27:131.
- Sande, Hannah, Peter Jenks, and Sharon Inkelas. 2020. Cophonologies by ph (r) ase. *Natural language & linguistic theory* 38:1211–1261.
- Selkirk, Elisabeth. 1982. The syllable. In *The Structure of Phonological Representations*, ed. Harry van der Hulst and Norval Smith, volume Part II. Dordrecht: Foris Publications.
- Selkirk, Elisabeth. 1995. The prosodic structure of function words. In *University of Massachusetts Occasional Papers: Papers in Optimality Theory*, ed. Jill Beckman, Laura Walsh Dickey, and Suzanne Urbanczyk, 439–470. Amherst, MA: GLSA Publications.
- Shetler, Joanne. 1976. *Notes on Balangao grammar*. Number 9 in Language Data: Asian-Pacific Series. Huntington Beach, CA: Summer Institute of Linguistics.
- Stanton, Juliet. 2016. Learnability shapes typology: the case of the midpoint pathology. *Language* 92:753–791.
- Staubs, Robert D. 2014. Computational modeling of learning biases in stress typology. Doctoral dissertation, UMass Amherst.
- Steriade, Donca. 1988. Reduplication and syllable transfer in Sanskrit and elsewhere. *Phonology* 5:73–155.
- Stevens, Alan M. 1968. *Madurese Phonology and Morphology*. New Haven, CT: American Oriental Society (American Oriental Series 52).
- Truckenbrodt, Hubert. 1999. On the relation between syntactic phrases and phonological phrases. *Linguistic Inquiry* 30:219–256.
- Urbanczyk, Suzanne. 2006. Reduplicative Form and the Root-Affix Asymmetry. *Natural Language & Linguistic Theory* 24:179–240.

- Wang, Yang. 2024. Studies in morphophonological copying: Analysis, experimentation and modeling. Doctoral dissertation, UCLA.
- Weeda, Donald. 1992. Word Truncation in Prosodic Morphology. Doctoral Dissertation, University of Texas, Austin.
- White, James, Rene Kager, Tal Linzen, Giorgos Markopoulos, Alexander Martin, Andrew Nevins, Sharon Peperkamp, Krisztina Polgárdi, Nina Topintzi, Ruben van De Vijver, et al. 2018. Preference for locality is affected by the prefix/suffix asymmetry: Evidence from artificial language learning. In *Proceedings of the Forty-Eighth Annual Meeting of the North East Linguistic Society (NELS 48)*, volume 3, 207–219. GLSA, Department of Linguistics, University of Massachusetts.
- Wiese, Richard. 2001. Regular morphology vs. prosodic morphology? The case of truncations in German. *Journal of Germanic Linguistics* 13:131–177.
- Wilson, Colin. 2006. Learning Phonology with Substantive Bias: An Experimental and Computational Study of Velar Palatalization. *Cognitive Science* 30:945–982.
- Wilson, Jennifer. 2014. Evidence for infixation after the first syllable: data from a Papuan language. *Phonology* 31:511–523.
- Wilson, Jennifer. 2017. A grammar of Yeri, a Torricelli language of Papua New Guinea. Doctoral dissertation, State University of New York at Buffalo.
- Wolf, Matthew. 2007. For an autosegmental theory of mutation. In *UMOP 32: Papers in Optimality Theory III*, ed. Leah Bateman, Michael O’Keefe, Ehren Reilly, and Adam Werle, volume 315–404. Amherst, MA: GLSA.
- Yu, Alan. 2003. The Morphology and Phonology of Infixation. Doctoral Dissertation, UC Berkeley.
- Yu, Alan CL. 2007. *A natural history of infixation*, volume 15. Oxford University Press.
- Zimmermann, Eva. 2021. Faded copies: Reduplication as distribution of activity. *Glossa* 58.
- Zoll, Cheryl. 1997. Conflicting directionality. *Phonology* 14:263–286.
- Zoll, Cheryl. 1998. Positional asymmetries and licensing. Ms., MIT. Available as ROA-282 on the Rutgers Optimality Archive, <http://roa.rutgers.edu>.
- Zukoff, Sam. 2023. Reduplicant shape alternations in Tawala: Re-evaluating base-dependence URL <https://ling.auf.net/lingbuzz/007412>.
- Zuraw, Kie, M Yu Kristine, and Robyn Orfitelli. 2014. The word-level prosody of samoan. *Phonology* 31:271–327.